RakeFlow - Research notes and data sources

Central Railway / Indian Railways freight operating statistics

Research date: 2026-10-13.

1. Average speed of freight trains

Figure	Value	Source
IR network average freight speed, 2013-24	**25.14 km/h** ("stagnant at ~25 km/h over the last 11 years") — Parliamentary Standing Committee on Railways report tabled 13 Dec 2024	https://www.maritimegateway.com/increasing-average-speed-of-freight-trains-is-of-prime-importance-report/ ; https://indiaseatradenews.com/rail-freight-in-india-slow-speeds-and-missed-opportunities/
IR network average freight speed, 2024-25	**23.8 km/h** ("due to prioritisation of passenger trains and ongoing works"); DFC average **37 km/h** — PRS summary of Standing Committee report	https://prsindia.org/files/policy/policy_committee_reports/Railways_enhancing_freight_revenue.pdf
Central Railway-specific average goods speed	**Not published** in the CR documents accessed (CR Highlights 2024-25, About Us Dec 2023). Use IR network values above.	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf

2. Wagon turnround (Nagpur division / CR)

Figure	Value	Source
Wagon turnround 2024-25 vs 2023-24	**1.46 wagon-days vs 1.54** (−5.2 %); wagon-km per wagon-day 187 vs 177 (+5.6 %); NTKM per wagon-day 7,409 vs 6,810 (+8.8 %)	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
Wagon turnround, FY25 to Jan-25	1.49 vs 1.69 wagon-days (−6.9 %)	https://cr.indianrailways.gov.in/cris//uploads/files/1740559903393-high%20lights.pdf
Freight interchange, Nagpur division	228.6 trains/day (vs 222.3, +2.8 %), FY25 to Jan-25; punctuality 80.55 %	same

(These two "Highlights" PDFs are hosted by CR and contain Nagpur-division-specific items — Tadali-Ghuggus caution orders, Majri–Tadali CRS, Hinganghat EI etc. — so the turnround figures are taken to be Nagpur division's; the documents do not say so explicitly.)

3. Detention / interchange delays (Nagpur area, Nov 2025)

Location	Trains/day	Detention / crossing time	Source
Ballarshah, interchange with SCR	~50	72 min	https://www.thehitavada.com/Encyc/2025/11/4/eyeing-3000-mt-loading-rlys-plans-to-end-detention-at-inter-changes.html
Ballarshah, interchange with SECR	~50	~207 min	same
Nagpur station end (CR)	up to 34	~204 min	same
Nagpur end (SECR side)	~35	~102 min	same
Godhni (with CR)	~26	~67 min	same
Kiratgarh (SCR)	~36	~30 min	same
Parasia (with SCR)	~7	~138 min	same
Pimpalkuti crossover to SCR	~5.5 rakes	n/a	same
Rail modal share	27–29 % (2024-25) vs road 65–70 %; target 45 % by 2030-31; 3,000 MT IR loading goal	same	

Detention at WCL sidings specifically: **not published**; FOIS FAQ gives demurrage Rs 150/wagon/hour and says free time varies by terminal (https://fois.indianrail.gov.in/RailSAHAY/pages/FAQs.jsp).

4. Loading volumes

Figure	Value	Source
Central Railway 2024-25	67.11 MT loading, Rs 7,217 crore (zonal)	https://indiaseatradenews.com/central-railway-records-freight-loading-of-67-11mt-fetches-%E2%82%B97217-crore-for-the-2024-25-fiscal/ ; https://indianinfrastructure.com/?p=207478
Central Railway April 2026	6.72 MT (+2 %), 2,582 rakes, Rs 708 crore; Nagpur 3.97 MT / Rs 432 crore; Mumbai 1.59; Solapur 0.59; Bhusaval 0.43; Pune 0.14	https://indianinfrastructure.com/2026/05/14/central-railway-achieves-freight-loading-of-6-72-mt-during-april-2026/
Nagpur division 2022-23	44.39 MT, >54 % of CR; coal ≈ 90 %; 668/958.70 BG route-km electrified; 94 block stations; 26 IBS	https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf
Nagpur division 2024-25	45.47 MT (vs 50.05 MT), cement 94 units/day (+18.5 %), clinker −138 rakes, iron ore −228 rakes, containers −207 rakes	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
Nagpur division monthly rake rate	May 2024: 1,142 rakes (≈37/day), coal 910 (≈29/day)	https://indiaseatradenews.com/central-railways-nagpur-division-records-freight-earnings-of-rs-427-cr-for-may/

5. Derived parameters for the simulation

- Tonnes per coal rake ≈ 2.14 MT / 536 rakes ≈ **3,990 t** (CR Highlights 2024-25).
- Section running speed: use 23.8–25 km/h network average as a conservative mean; allow 30–40 km/h on a short, light-traffic spur as an optimistic case.
- Wagon turnround 1.46 days is a system-level KPI for validating an extended (Ghugus → power plant → return) model, not the 14 km section alone.
- Interchange/junction detentions of 1–3.5 h at Ballarshah/Nagpur indicate the main-line side can block departures from Tadali; model main-line slot availability as a stochastic delay (mean ~1–2 h).

CSTPS (Chandrapur STPS, MAHAGENCO) – coal demand, stock and receipt data

Research date: 2026-10-13. All figures are from the CEA "Daily Coal Stock Report" PDFs hosted on the National Power Portal (npp.gov.in), unless stated otherwise. CSTPS appears as "CHANDRAPUR (MAHARASHTRA) STPS", Sl. No. 31 (30 in 2024 reports), mode of transport = RAIL, capacity 2920 MW (2x210 + 5x500; units 1-2 retired, units 3-9 operating).

Source URL pattern (dd-mm-yyyy): `https://npp.gov.in/public-reports/cea/daily/fuel/DD-MM-YYYY/dailyCoal1-YYYY-MM-DD.pdf` Index page: https://npp.gov.in (Reports -> CEA daily fuel reports). Note: direct download via curl is blocked by robots/proxy from our environment; open in browser.

1. Column definitions in the CEA daily report

- Capacity (MW)
- Normative Stock Reqd. (Days) – CEA norm; varies month to month (20-26 days observed; pithead/rail-fed norm)
- Daily Requirement @85% PLF ('000 T) – this is CEA's *normative* consumption, NOT actual
- Actual Stock ('000 T): Indigenous / Import / Total
- % of Actual Stock vis-a-vis Normative Stock; plant flagged "Critical (*)" when < 25% of normative
- Receipt of the day ('000 T) <-- key input for the simulation (coal delivered that day)
- Consumption of the day ('000 T) <-- actual burn
- Current month PLF (tentative)

2. CSTPS values by report date (all tonnes in '000 T)

Report date	Norm. days	Daily req @85% PLF	Stock indig.	Stock import	Stock total	% of norm	Receipt of day	Consumption of day	Month PLF	URL
25-07-2024	22	47.2	617.5	157.6	775.1	75%	17.3	18.7	56%	https://npp.gov.in/public-reports/cea/daily/fuel/25-07-2024/dailyCoal1-2024-07-25.pdf
15-10-2025	21	49.5	233.3	46.1	279.5	27%	29.9	32.6	58%	https://npp.gov.in/public-reports/cea/daily/fuel/15-10-2025/dailyCoal1-2025-10-15.pdf
08-12-2025	23	49.5	740.3	52.3	792.6	70%	3.9	32.1	50%	https://npp.gov.in/public-reports/cea/daily/fuel/08-12-2025/dailyCoal1-2025-12-08.pdf
02-06-2026	26	47.8	475.6	114.2	589.8	47%	47.5	31.5	57%	https://npp.gov.in/public-reports/cea/daily/fuel/02-06-2026/dailyCoal1-2026-06-02.pdf
14-06-2026	26	47.8	580.8	131.9	712.7	57%	33.8	34.9	58%	https://npp.gov.in/public-reports/cea/daily/fuel/14-06-2026/dailyCoal1-2026-06-14.pdf
17-07-2026	22	47.8	724.6	80.9	805.5	77%	27.1	32.9	54%	https://npp.gov.in/public-reports/cea/daily/fuel/17-07-2026/dailyCoal1-2026-07-17.pdf
31-07-2026	22	47.8	726.3	79.1	805.5	77%	16.2	22.3	51%	https://npp.gov.in/public-reports/cea/daily/fuel/31-07-2026/dailyCoal1-2026-07-31.pdf
02-09-2026	20	47.8	478.6	9.1	487.7	51%	31.1	43.6	71%	https://npp.gov.in/public-reports/cea/daily/fuel/02-09-2026/dailyCoal1-2026-09-02.pdf
05-09-2026	20	47.8	455.4	3.4	458.8	48%	37.5	47.8	40%(?)	https://npp.gov.in/public-reports/cea/daily/fuel/05-09-2026/dailyCoal1-2026-09-05.pdf

Caveats: values were extracted from PDF text (column layout scrambles in text extraction). Indigenous+Import=Total reconciles in every row, so stock figures are reliable; receipt/consumption/PLF were mapped by column position and should be spot-checked visually against the PDF before being quoted in a report. The 05-09-2026 PLF (40%) looks inconsistent with consumption 47.8 kt and is probably mis-mapped.

Derived numbers for the simulation

- CEA normative requirement at 85% PLF: **47.8 kt/day** (2026), 49.5 kt/day (late 2025), 47.2 (2024). Implied specific coal consumption: 47,800 t / (2920 MW x 0.85 x 24 h) = ~0.80 kg/kWh (typical for ~3,600-3,800 kcal/kg WCL coal).
- Actual consumption observed: 18.7 - 47.8 kt/day, median of the 9 observations ~32.6 kt/day (PLF 50-58%).
- Actual receipt observed: 3.9 - 47.5 kt/day, median ~29.9 kt/day. Receipt includes ALL modes (rail + ropeway + pipe conveyor + road).
- Stock on hand: 280 - 806 kt => 6 - 17 days of normative requirement.
- Normative stock: 20-26 days x 47.8 kt = 957 - 1,244 kt (CEA changes the day-norm seasonally).

- Nowhere in the reports is a "rakes per day" column; convert: receipt_kt / ~3.9 kt per rake (see rake_payload_norms.md). E.g. 31.1 kt/day => ~8 rakes-equivalent; 47.5 kt/day => ~12 rakes-equivalent. Subtract ~7 kt/day arriving by ropeway/pipe conveyor (see below) to get the rail share: roughly **6 - 10 rakes/day by rail** on normal days, with bursts to ~12. This is consistent with the "8-14 rakes/day" prior for a 2920 MW plant only at high PLF.

3. MAHAGENCO internal figure for daily requirement

CPRI vetting report for MAHAGENCO (Coal handling plants at CSTPS), hosted on mahagenco.in: https://www.mahagenco.in/uploade/Petition/Data%20Gaps%20Set-I%20Annexures/DG%20-%20026%20_Vetting%20Reports/CPRI%20Vetting%20REPORT.pdf

- Sec 1.1: "The daily requirement of coal is @ 47000 MT for overall CSTPS."
- CHP-A (units 3-4, 2x210 MW): daily requirement ~8,000 MT; receives coal by Rail, Rope and Road; ~7,000 MT/day comes by the 1.2 km aerial ropeway from WCL Durgapur mine (Sec 4.12).
- CHP-B (units 5-7, 3x500 MW): daily requirement ~24,000 MT; rail (BOXN and BOBR wagons).
- CHP-C: track hopper for BOBR/UTS wagons, road receipt facility, and a closed pipe conveyor from WCL Bhatadi OCM.
- CHP-A/B/C combined handle ~32,000 MT/day for 2x210 + 3x500 MW (Sec 4.11).
- CHP-D (units 8-9, 2x500 MW): remainder (~15,000 MT/day implied), rail via 3 wagon tipplers.

4. Coal sources and modes (rail vs road vs conveyor; no MGR)

- MoEF&CC six-monthly EC compliance report Jan-Jun 2025 (MAHAGENCO): https://mahagenco.in/clearance_pdf/MoEF&CC_Six%20Monthly%20EC%20Compliance%20report%20for%20period%20of%20January%202025%20-%20June%202025%20for%20CSTPS%20Chandrapur_20250724120519827.pdf
  - Coal is from Western Coalfields Ltd (WCL); mines named around the plant: Bhatadi, Durgapur, Padmapur (WCL Chandrapur Area). CSTPS has a linkage to Durgapur OC mine.
  - Modes: rail wagons, aerial ropeway (Durgapur), closed pipe conveyor from Bhatadi/Padmapur mine area, trucks (road).
  - No quantities by mode are given. No merry-go-round (MGR) system exists at CSTPS (none mentioned in any MAHAGENCO/CPRI/EC document; the Wikipedia page lists Durgapur and Padmapur collieries as sources).
- Wikipedia: https://en.wikipedia.org/wiki/Chandrapur_Super_Thermal_Power_Station (coal from WCL Durgapur and Padmapur).
- Units 8&9 EC (MoEF letter J-13011/53/2008-IA.II(T), 30-01-2009, extended 31-03-2016): coal source stated as the Machakata coal block (Odisha; never developed) – in practice units 8-9 run on WCL linkage + imports. https://mahagenco.in/Content/File_pdf/EnvClearance/5.%20CSTPS%20U%208%20&%209%20EC%20&%20extension,%20ammendments.pdf
- CEA daily reports show a persistent "Import" stock component at CSTPS (3 - 158 kt), so imported/blended coal also arrives, by rail from ports.

- Rail-side origin (Central Railway, Nagpur Division): WCL loading sidings at Ghugus, Wani, Umrer, Majri, Ballarpur areas. Record-day data for Nagpur Division, 5-Nov-2020: 42 coal rakes loaded in the division in one day (Ghugus 16 rakes, Umrer 11, Wani 6); Oct-2020 division loading 3.28 MT. https://news.railanalysis.com/?p=37597
- Central Railway Highlights 2024-25: coal loading from WCL sidings fell by 1.5 rakes/day (536 rakes, -2.14 MT), implying ~4.0 kt/rake on CR coal loading. https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
- Tadali-Ghugus (14 km, Wardha-Ballarshah HDN route) is single line; doubling approved July 2026, expected to add 4.5 MTPA freight. https://indianinfrastructure.com/2026/07/22/ir-approves-tadali-ghugus-doubling-rail-line-project-in-maharashtra/ This is the critical section for rakes from Ghugus/Wani to CSTPS and to Dhariwal (Tadali).

5. Dhariwal Infrastructure (CESC), Tadali, 2x300 MW – competing user of Tadali

- CEA daily report lists it as "DHARIWAL TPP", IPP section, mode RAIL, 600 MW:
  - 17-07-2026: daily req @85% PLF 7.8 kt; stock 137.8 kt; receipt 5.8 kt; consumption 8.5 kt.
  - 05-09-2026: daily req 7.8 kt; stock 98.2 kt; receipt 5.1 kt; consumption 8.8 kt. => ~1.5 - 2.5 rakes/day (approx 2 rakes/day at ~3.9 kt/rake).
- Coal: FSA with **SECL** (not WCL) operational since March 2016, 1.58 MTPA for unit 2 (long-term PPAs); 1.15 MTPA under SHAKTI for unit 1 medium-term PPAs; shortfall via e-auction/imports. ICRA rationale 14-Sep-2026: https://www.icra.in/Rating/ShowRationalReportFilePdfViewer/145562 ; Infomerics PR 26-Aug-2016: https://infomericstorage.blob.core.windows.net/uploads/pr_Dhariwal_Infrastructure_feb16_528e229a5b.pdf
- Since its coal comes mostly from SECL (Chhattisgarh) it arrives on the Nagpur-Ballarshah trunk from the north, i.e. it uses the Tadali station/siding but does NOT use the Tadali-Ghugus single-line branch; it does compete for Tadali yard capacity and for trunk-line paths Wardha-Tadali. Wikipedia: https://en.wikipedia.org/wiki/Dhariwal_Power_Station
- GMR Warora (600 MW, Warora, also Nagpur division, flagged critical 05-09-2026: stock 36.4 kt, req 7.9 kt/day) is another WCL-fed plant on the Wardha-Ballarshah route.

6. Other data sources checked

- MERIT (meritindia.in) – gives merit-order/variable cost, not coal logistics; not used.
- MAHAGENCO website – plant pages have no daily coal data; the EC compliance reports (links above) and the CPRI vetting report are the useful documents. A list of all CSTPS EC/compliance PDFs is at https://mahagenco.in/Environmental-Clearance

Tadali–Ghugus doubling project (sanctioned July 2026)

Research date: 2026-10-13.

1. Sanction and headline figures

Item	Value	Source
Approval	Approved by Indian Railways "on Thursday" (16 July 2026, per RailPost article dated 17 July 2026); sanctioned under "umbrella works for doubling/tripling/quadrupling/flyover/bypass works"	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/
Completion cost	**Rs 228 crore**	https://kashmirdespatch.com/14-km-doubling-project-to-enhance-freight-capacity-and-improve-operational-efficiency-on-delhi-chennai-high-density-network/ ; https://www.itln.in/railway/indian-railways-doubles-tadalighugus-line-for-228-cr-1359929 ; https://www.nbmcw.com/news/metro-railways/indian-railways-approves-doubling-of-tadali-ghugus-railway-section-in-maharashtra-at-rs-228-cr.html
Length	**14 km** (14.11 km per Gazette)	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/ ; https://egazette.gov.in/WriteReadData/2026/273346.pdf
Additional traffic expected	**4.5 million tonnes per annum (MTPA)** on completion	https://www.drishtiias.com/state-pcs-current-affairs/indian-railways-approves-tadali-ghugus-railway-line-doubling-project-in-maharashtra/print_manually ; https://www.itln.in/railway/indian-railways-doubles-tadalighugus-line-for-228-cr-1359929
Commodities named	Coal, iron ore, gypsum, IOBT POL (petroleum products), sand, laterite ash	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/
Stated justification	"Presently single-line"; "growing freight demand"; "add much-needed capacity as line capacity grows, improve train operations"; "congestion on the corridor … lower logistics costs for bulk cargo"	https://kashmirdespatch.com/… (above) ; https://www.itln.in/railway/indian-railways-doubles-tadalighugus-line-for-228-cr-1359929
Route classification	Spur off the Wardha–Balharshah section of the **Delhi–Chennai High Density Network (HDN)**; CR Nagpur Division	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/
Land acquisition	Gazette notification dated 4 June 2026 (i.e., before the July sanction news) declaring the 14.11 km doubling a "Special Railway Project" under Section 37A for land acquisition in Chandrapur district	https://egazette.gov.in/WriteReadData/2026/273346.pdf
Expected completion	**Not stated in any source found.**	—
Current line-capacity utilisation (%) / trains per day on the spur	**Not stated in any source found.** All articles (RailPost, ITLN, NBM&CW, Drishti, Kashmir Despatch, Indian Infrastructure, Central Chronicle, DD India) carry the same PIB-style text with only the 4.5 MTPA figure.	https://indianinfrastructure.com/2026/07/22/ir-approves-tadali-ghugus-doubling-rail-line-project-in-maharashtra/ ; https://centralchronicle.in/indian-railways-approves-%E2%82%B9228-crore-doubling-of-tadali-ghugus-rail-section-in-maharashtra/ ; https://ddindia.co.in/2026/07/indian-railways-approves-%E2%82%B9228-crore-doubling-of-tadali-ghugus-rail-section-in-maharashtra/

Derived: 4.5 MTPA extra ≈ 4.5e6 / (~3,990 t per coal rake, see wcl file) ≈ 1,130 rakes/yr ≈ **3.1 additional loaded rakes/day** (own calculation; useful as a capacity-gain target for the simulation).

2. Related corridor projects (Wardha–Balharshah / Wardha–Nagpur)

Project	Figures	Source
Wardha (Sewagram)–Balharshah 3rd line	132.34 km; Rs 1,384.72 crore; 67 % progress (Mar 2025). Originally sanctioned 2015-16 as "Wardha (Sewagram)-Balharshah 3rd line, 132 km". 2022 article quoted 160 km / Rs 1,035.16 crore / ~25 % complete.	https://cr.indianrailways.gov.in/cris//uploads/files/1742449942265-List_of%20Projects_%20Progress_%20upto_Mar2025.pdf ; https://eparlib.nic.in/bitstream/123456789/774300/1/AU740.pdf ; https://www.thehitavada.com/Encyc/2022/3/12/No-solid-progress-in-Rly-projects-aimed-at-decongesting-clogged-network.html
3rd line sections opened	Chitoda–Sonegaon, Sonegaon–Hinganghat, Sewagram–Chitoda, Chikni Road–Majri, Tadali–Ballarshah (May 2025, >80 % done); Majri–Tadali CRS inspection 10.01.2025	https://www.thehitavada.com/Encyc/2025/5/29/Union-Cabinet-gives-nod-to-4th-line-on-Wardha-Ballarshah-route.html ; https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
Wardha–Balharshah 4th line	Union Cabinet approval 28 May 2025; Rs 2,318 crore; completion target 2029-30; existing route 132 km; (with Ratlam–Nagda) adds ~176 km, Rs 3,399 crore total, +18.40 MT/yr freight, 784 villages/19.74 lakh population, 20 crore litres oil saved, 99 crore kg CO2	https://www.thehitavada.com/Encyc/2025/5/29/Union-Cabinet-gives-nod-to-4th-line-on-Wardha-Ballarshah-route.html
Wardha–Nagpur 3rd line	76.23 km; Rs 540.02 crore; 72 % (Mar 2025)	https://cr.indianrailways.gov.in/cris//uploads/files/1742449942265-List_of%20Projects_%20Progress_%20upto_Mar2025.pdf
Wardha–Nagpur 4th line	78.71 km; Rs 637.95 crore; 70 % (Mar 2025); sanctioned 2016-17 (76 km)	same as above; https://eparlib.nic.in/bitstream/123456789/774300/1/AU740.pdf
Nagpur–Itarsi 3rd line	279.28 km; Rs 2,449.91 crore; 48 % (Mar 2025)	https://cr.indianrailways.gov.in/cris//uploads/files/1742449942265-List_of%20Projects_%20Progress_%20upto_Mar2025.pdf
Congestion context (Nov 2025)	Ballarshah interchange with SCR ~50 trains/day, exchange lag 72 min; with SECR ~207 min; Nagpur end up to 34 trains, ~204 min; RoR proposed at Majri Jn; Ballarshah bypass, extra loops, Chandrapur–Chanda Fort chord proposed; Wardha–Ballarshah 4th line DPR under preparation	https://www.thehitavada.com/Encyc/2025/11/4/eyeing-3000-mt-loading-rlys-plans-to-end-detention-at-inter-changes.html
Tadali goods-shed works (Jan 2025)	Rs 26.94 crore for a 4th line and goods-shed platform at Tadali; Ghugus goods shed listed for improvement (no amount)	https://www.thehitavada.com/Encyc/2025/1/25/Rs-100-crore-works-ayed-for-upgradation-of-goods-shed.html

3. Implications for the simulation

- Baseline scenario: one single-line block Tadali–Ghugus. Doubling scenario: two independent block lines (one per direction), same terminals.
- The officially expected gain is +4.5 MTPA ≈ +3 loaded rakes/day; the model's capacity-gain estimate can be benchmarked against this.
- No official current throughput figure for the spur exists in public sources; WCL siding figures (see wcl_ghugus_loading.md) are the best proxy.

Ghugus–Tadali section: infrastructure facts

Research date: 2026-10-13. Every figure is tagged with its source URL. Items marked **[NOT VERIFIED]** could not be confirmed from an accessible source and should be treated as modelling assumptions.

1. Identity and codes

Item	Value	Source
Ghugus station code	**GGS** (not "GUGS"), Central Railway (CR), Nagpur (NGP) division, Chandrapur district, Maharashtra, broad gauge; "goods station", traffic type "all"	https://erail.in/stations/GGS
Ghugus spelling in CR documents	CR's own Highlights 2024-25 spells it "Ghuggus" ("Tadali-Ghuggus section")	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
Tadali station code	**TAE**, Tadali Jn, Nagpur division, on New Delhi–Chennai main line; "a line to Ghugus branches off to the south-west"	https://en.wikipedia.org/wiki/Tadali ; https://erail.in/stations/TAE
Tadali station category	E category (Nagpur CR division station list)	https://en.wikipedia.org/wiki/Nagpur_CR_railway_division
Ghugus passenger status	Listed under "stations closed for passengers"; end point of the "Tadali Jn to Ghugus" branch	https://en.wikipedia.org/wiki/Nagpur_CR_railway_division
Ghugus town–Chandrapur distance	~20.92 km (13 mi); branch line "connects Ghugus to the Chandrapur–Wardha main line … branching off at Tadali"	https://en.wikipedia.org/wiki/Ghugus
Passenger trains Ghugus–Tadali today	None. erail lists no trains at GGS; rome2rio: "There is no direct connection from Wardha to Ghugus" (nearest practical station Bhandak + taxi)	https://erail.in/stations/GGS ; https://www.rome2rio.com/s/Wardha/Ghugus

Note on the "Wardha–Ghugus passenger": no source was found confirming such a service ever ran. The Lok Sabha 1992 record found in search (https://eparlib.nic.in/bitstream/123456789/1097397/1/10_V_22-12-1992_p82_p83_u4705.pdf) does not mention Ghugus. **[NOT VERIFIED]**

2. Line length and geometry

Item	Value	Source
Length Tadali–Ghugus	**14.11 km** (Gazette of India notification, Ministry of Railways/Central Railway, 4 June 2026, declaring "New Double line between Tadali–Ghugus Doubling Project (14.11 Kms)" a Special Railway Project under Sec. 37A, Railways (Amendment) Act 2008, for land acquisition in Chandrapur district)	https://egazette.gov.in/WriteReadData/2026/273346.pdf
Length (rounded)	14 km, "single-line spur on the Wardha–Balharshah section"	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/
Intermediate block stations	None found in any station list. Nagpur CR division list shows the branch simply as "Tadali Jn to Ghugus" with no intermediate station. **Treat as one block section Tadali–Ghugus [NOT VERIFIED beyond absence in lists]**	https://en.wikipedia.org/wiki/Nagpur_CR_railway_division

Item	Value	Source
Gradient / ruling grade	Not published in any accessible source. **[NOT VERIFIED]** The 2005 MoEF document for Ghugus OCP gives no track data.	—
Sectional speed (MPS)	Not published. **[NOT VERIFIED]** Branch is goods-only; for simulation assume 30–50 km/h with freight average ~25 km/h network-wide (see cr_freight_statistics.md).	—

3. Track configuration, block working, electrification

Item	Value	Source
Single/double	**Single line** today ("The section is presently single-line")	https://kashmirdespatch.com/14-km-doubling-project-to-enhance-freight-capacity-and-improve-operational-efficiency-on-delhi-chennai-high-density-network/
Block working	Not stated. CR Highlights 2024-25 says ICMS-based real-time caution orders (COA) were implemented on "Tadali-Ghuggus sections" from 15 January 2025, implying the section is on the divisional control/ICMS system. Token vs tokenless **[NOT VERIFIED]**; Nagpur division reports 26 IBS (intermediate block signalling) locations and 94 block stations division-wide (Dec 2023)	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf ; https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf
Electrification	Central Railway completed 100 % electrification of its entire 3,825 km broad-gauge network on 23 Feb 2023 (last section Ausa Road–Latur Road, 52 km). Hence Tadali–Ghugus (BG, CR) is electrified. Nagpur division had 668 of 958.70 BG route km electrified as of Dec 2023 — the remainder is presumably the Wardha–Nanded new line etc.	https://www.railjournal.com/infrastructure/indias-central-railway-completes-electrification-of-its-entire-network/ ; https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf
Number of loops at Ghugus	Not published. erail shows two sidings taking off from GGS: **GSG "Ghugus Colliery Siding" (4 km)** and **CESG "Associated Cement Co. Siding – Ghugus" (2 km)**. **[loops NOT VERIFIED]**	https://erail.in/stations/GGS ; https://vcrc.in/wp-content/uploads/2020/05/List_Coal_Siding_090719.pdf
Number of loops at Tadali	Not published. Tadali has a major goods shed, POL unloading (IOBT siding for IOC/BPCL), PVIT (Vimla Infrastructure) coal siding and MDIT (Dhariwal) private power-house siding; Rs 26.94 crore sanctioned (Jan 2025) for a "fourth line and a goods shed on platform 4 (16,000 sq m)" at Tadali	https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf ; https://www.thehitavada.com/Encyc/2025/1/25/Rs-100-crore-works-ayed-for-upgradation-of-goods-shed.html ; https://vcrc.in/wp-content/uploads/2020/05/List_Coal_Siding_090719.pdf

4. Main line context at Tadali (Wardha–Balharshah)

Item	Value	Source
Wardha–Balharshah route	Part of Delhi–Chennai High Density Network (HDN); double line with 3rd line under construction	https://www.railpost.in/ir-approves-14-km-tadali-ghugus-section-doubling-at-a-cost-of-228-crore/
Sewagram–Balharshah 3rd line	132.34 km, Rs 1,384.72 crore, 67 % progress as of March 2025	https://cr.indianrailways.gov.in/cris//uploads/files/1742449942265-List_of%20Projects_%20Progress_%20upto_Mar2025.pdf
Majri–Tadali 3rd line	CRS inspection on 10.01.2025	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf

Item	Value	Source
Sections of 3rd line already open (May 2025)	Chitoda–Sonegaon, Sonegaon–Hinganghat, Sewagram–Chitoda, Chikni Road–Majri, Tadali–Ballarshah; >80 % physical work complete	https://www.thehitavada.com/Encyc/2025/5/29/Union-Cabinet-gives-nod-to-4th-line-on-Wardha-Ballarshah-route.html
Stations Wardha→Balharshah (division list)	Wardha Jn, Majri Jn, Tadali Jn, Chandrapur, Balharshah Jn (list is not exhaustive of halts)	https://en.wikipedia.org/wiki/Nagpur_CR_railway_division

5. Suggested simulation parameters derived from the above

- One single-line block section, Tadali ↔ Ghugus, length 14.11 km.
- Terminal at Ghugus: two sidings (GSG colliery siding, 4 km; CESG ACC cement siding, 2 km). Model each as a loading berth with its own placement/release times.
- Terminal at Tadali: junction with the double (becoming triple) Wardha–Balharshah main line; empties arrive from and loads depart to the main line.
- Loop counts at both ends are unknown — treat as a sensitivity parameter (1–3 loops at Ghugus, 2–4 at Tadali).
- Running time per block occupation: 14.11 km at 25–40 km/h ≈ 21–34 min, plus block clearance/signalling overhead.

Indian Railways freight statistics, terminal detention distributions, and single-line capacity

Purpose: system-level validation targets (wagon turnround, speed) and block-section / crossing parameters for the coal rake-cycle DES (Central Railway, single-line sections ~14 km).

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1. Official all-India freight indices

Source: Indian Railways Annual Report & Accounts 2023-24 (Railway Board, Statistics & Economics Dte), pp. 6, 13, 15. https://indianrailways.gov.in/railwayboard/uploads/directorate/stat_econ/2025/Indian%20Railways%20Annual%20Report%20%20Accounts%202023-24%20-English.pdf

Index (Broad Gauge)	2022-23	2023-24
Wagon turnround (days)	4.70	**5.11**
Average speed of goods trains (km/h, all traction)	30.80 (rev.)	**25.00**
NTKM per wagon-day	8,670	7,851

Index (Broad Gauge)	2022-23	2023-24
Wagon-km per wagon-day	223	192
Net load per goods train (t)	1,738	1,899
Average load per wagon (t)	60	65
Average lead of revenue freight (km)	636	613

Other years:

- 2021-22: NTKM/wagon-day 8,384; average lead 615 km (Railway Board statistical summary sheet 2021-22): https://indianrailways.gov.in/railwayboard/uploads/directorate/stat_econ/2024/REVISED%20Summery%20Sheet%202021-22%20(English).pdf
- 2014-15: WTR 5.6 days; goods average speed 25 km/h; lead 622 km (AITD training slides "Freight Operations & Operating Indices"; defines WTR = effective wagon holding / (loaded wagons + loaded receipts) and speed = train-km / train-engine-h): https://mail.aitd.net.in/ppt/21/5.%20Freight%20Operations%20&%20Operating%20Indices.pdf
- 2019-20: goods average 23.6 km/h vs Mission Raftaar target 50 km/h (PAC 36th Report, 18th Lok Sabha, para 14, following C&AG Report 22 of 2021): https://eparlib.sansad.in/bitstream/123456789/3017115/1/18_Public_Accounts_36.pdf
- May 2021 (Covid low-traffic): WTR 4.83 days (vs 6.61 in May 2019), speed 45.42 km/h (PIB via Rail Analysis): https://news.railanalysis.com/?p=41408
- 2024-25: average freight speed **23.8 km/h**; DFC 37 km/h (Standing Committee on Railways report of 16.12.2025, PRS summary): https://prsindia.org/files/policy/policy_committee_reports/Railways_enhancing_freight_revenue.pdf
- Standing Committee (Dec 2024): 25.14 km/h, stagnant ~11 years: https://indiaseatradenews.com/rail-freight-in-india-slow-speeds-and-missed-opportunities/
- CAG Report 31 of 2014 para 3.2.4: speed "almost static at around 25 km" 2008-13; FOIS Feb 2012 sample of 6,730 loaded trains in 7 zones: **71% averaged <= 20 km/h, 17% 20-40 km/h, 12% 40-100 km/h**; permissible 60 km/h. All-India WTR 5.19 -> 5.08 days (2008-13). https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf

Coal-chain specifics (BigMint, secondary): agreed rake turnaround 48-72 h vs summer actual 96-120 h; colliery-to-trunk-route segment 12-24 h; Grand Chord utilisation 120-130% in summer; EDFC speeds 45-50 km/h vs 25-30 km/h on conventional lines. https://www.bigmint.co/insights/detail/indias-coal-supply-chain-strong-on-output-weak-on-delivery-747729

2. Terminal and en-route detention distributions (CAG Report 31 of 2014)

Source: https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf Executive summary: https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_5_Exe_Summ.pdf

- Para 3.2.5: at 128 loading/unloading points (2008-13) **32% of wagons detained

  24 h**; 30% (loading) and 21% (unloading) of wagons suffered detention beyond free time in 2011-13; average detention during loading 12-86 h, during unloading 13-141 h; Railway Board's envisaged placement-to-release envelope **16 h** not achieved.

- Table 18 stage-wise zonal averages: loading arrival->placement 0:32-10:15; placement->release 5:03-56:32; release->despatch 0:49-21:03; unloading arrival->placement 0:38-15:07; placement->release 9:27-124:12; release->despatch 1:28-11:04.
- Para 3.2.6 yard detention norms (zone-set): ECoR 1 h, ECR 8 h, WR 10 h, **CR 15 h**, ER 24 h; actual exceeded norms by 1.46-36.53 h.
- Para 3.2.4 en-route: NWR rake detained 14 h 25 m, 442 km journey took 21 h; ER 11 rakes on the same route varied by 0-89 h; NR 32 trains detained > 24 h.
- Para 3.4: 63% of late starts for want of loco, 18% for crew/path.
- PAC 36th Report Table 2.8 (2018-19) goods detention at interchange points: NKJ 171-262 min, BHC 151-248 min, ITR 66-186 min, Gudur 2-213 min; six internal causes = 66% of detentions; line-capacity utilisation 137-175% on NCR/ECR trunk sections. https://eparlib.sansad.in/bitstream/123456789/3017115/1/18_Public_Accounts_36.pdf

3. Single-line block section working and line capacity

3.1 Scott's formula (Indian Railways standard)

C = (1440 / (T + t)) x E

- C = trains per day (both directions on single line); 1440 = minutes/day; **T = running time of the slowest train over the critical (longest-time) block section**; **t = block operation time** (line-clear, signal, points, token); **E = efficiency factor**, conventionally **0.70 for single line, 0.80 for double line**. IR practice deducts a maintenance block first: 1440 - 120 = 1320 min.
  - Formula as written in Sangphong, Siridhara & Ratanavaraha, Engineering Journal 21(3), 2017, eq. 1: https://engj.eng.chula.ac.th/ej/article/download/1266/580
  - IR recipe (1,320 min / (slowest-train running time + section preparation time) x 70%), "Who is Scott?", Business Standard 05.10.2017: https://business-standard.com/amp/article/opinion/who-is-scott-117100501278_1.html
  - Departmental exam note (T = running time of slowest train, t = block operation time, E = efficiency): https://mycbt.in/posts/knowledge-booster-advance-operations-edition-12-jul-2026
  - Same structure in JICA/Nippon Foundation single-track formula N = 1440 f / (T + C): block handling time C = **2 min 30 s (tablet/token)**, **1 min 30 s (automatic)**; utilisation f 0.5-0.75: https://nippon.zaidan.info/seikabutsu/1996/00501/contents/096.htm
- IR 2015 White Paper: 492 of 1,219 sections (40%) above 100% utilisation; block sections typically 10-15 km outside urban areas (Business Standard, above).

3.2 Block operation time t (for the model)

No single Railway Board number; the values in use in IR line-capacity work: tokenless block / panel interlocking **3-4 min**, token (Neale's) **5 min**, automatic block **1.5-2 min**. The Nippon Foundation JICA figures (2.5 min tablet, 1.5 min auto) bracket the low end; IR practice adds signal-clearance and reception time, so use tri(3, 4, 6) min for a tokenless single-line block with a crossing station at each end. General Rules ch. 14 (block working, line clear only after whole opposing train is inside and clear of running line, GR 14.11(4)): https://irfca.org/docs/rulebook/gr-block.html

3.3 Time to clear a 14 km single-line block

Pure running time at constant speed: 14 km / 50 km/h = **16.8 min**; / 40 = **21.0 min**; / 30 = **28.0 min**. Add start-from-stop and stop-at-loop allowances used in IR working timetables (typically 2-3 min each for a loaded 4,000-5,000 t BOXN train; illustrative station working rules: ECoR SWR documents https://eastcoastrail.indianrailways.gov.in/uploads/files/1719289299569-KTGA_FULL_SWR_01042024.pdf) and block operation t = 3-6 min:

Case	Occupancy of block (min)
Loaded coal rake, 30 km/h sectional, stop-to-stop	28 + 3 + 3 + 4 = **38**
Loaded, 40 km/h	21 + 3 + 3 + 4 = **31**
Empty BOXN, 50 km/h	17 + 2 + 2 + 4 = **25**

Scott capacity of the 14 km block: with T = 31 min, t = 4 min, E = 0.7 -> 1320/35 x 0.7 = **26 trains/day**; with T = 38 -> **22 trains/day**.

3.4 Crossing / precedence at a loop station

Under absolute block on single line, a train waiting to cross must stand on the loop until the opposing train has arrived complete and is clear (GR 14.11(4)). The crossing delay for the waiting train = (time from its arrival until opposing train arrives) + (block operation t) + (start allowance). Observed single-line crossing detentions in IR operating studies are typically 10-30 min per crossing; IIMA WP 1992-04-01 studied a 43 km coal/cement section where cross movements caused detention (no figures in abstract): https://ideas.repec.org/p/iim/iimawp/wp01096.html Model as tri(5, 15, 40) min per crossing for a goods train giving precedence.

4. Suggested validation targets and parameters

Quantity	Value	Basis
All-India WTR	4.7-5.6 days (2014-15 to 2023-24)	Annual Report 2023-24; AITD
Short-lead coal circuit turnaround (CIL-Railway agreed)	48-72 h; summer 96-120 h	BigMint

Quantity	Value	Basis
Freight average speed (incl. stops)	tri(18, 25, 32) km/h	Annual Report 25.0; CAG 71% <= 20 km/h; 2022-23 30.8
Sectional running speed, loaded BOXN on single line	tri(30, 40, 50) km/h	CAG permissible 60; EDFC 45-50
Block operation time t	tri(3, 4, 6) min	JICA 1.5-2.5 min + IR signalling allowances
Crossing wait at loop	tri(5, 15, 40) min	GR 14.11; IR operating practice
Terminal placement->release beyond free time (prob.)	21% (unloading), 30% (loading)	CAG para 3.2.5
Share of rakes detained > 24 h at terminals	32% (2008-13 IR-wide; use as pessimistic bound)	CAG para 3.2.5
Yard detention norm, CR	15 h	CAG para 3.2.6

Indian Railways norms: Brake Power Certificate (BPC) and freight-rake examination

Purpose: parameters for the "BPC validity / intensive examination" events in the coal rake-cycle DES (closed-circuit BOXN rakes). Primary source is the Railway Board Joint Procedure Order (JPO) on freight train examination as reproduced in IRIMEE (Indian Railways Institute of Mechanical & Electrical Engineering) course notes; the JPO derives from Railway Board (Mechanical) letters cited below. IRCA Conference Rules Part III govern interchange/fitness but the time standards used in practice are these JPO figures.

Primary sources:

- IRIMEE "Pattern of freight train examination" (JPO summary): https://rskr.irimee.in/sites/default/files/Pattern%20of%20freight%20train%20examination.pdf
- IRIMEE "Freight Train examination" (JPO): https://rskr.irimee.in/sites/default/files/Freight%20Train%20examination.pdf
- IRIMEE BPC note: https://rskr.irimee.in/sites/default/files/BPC.pdf
- IRIMEE C&W refresher material 16-10-2018: https://rskr.irimee.in/sites/default/files/RC%20C&W%20material%2016-10-2018.pdf
- SECR Zonal Safety Circular 05/2022 (BPC): https://secr.indianrailways.gov.in/uploads/files/1660131557781-Zonal%20Safety%20Circular%20No.%2005-2022.pdf

Railway Board letters cited in the JPO: 94/M(N)/951/57 Vol-II dated 25.10.2004 (CC rake formation, revised JPO basis); 2005/M(N)/951/13 dated 07.04.2005; even-number letters dated 25.04.2006 and 05.11.2007. https://rskr.irimee.in/sites/default/files/Pattern%20of%20freight%20train%20examination.pdf

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1. BPC validity by examination pattern

Pattern	Validity	Grace / notes	Source
Closed-circuit (CC) rake examined at "A" category depot	**6,000 km or 30 days**, whichever earlier	5-day grace if rake is moving toward base depot; if km not logged continuously, BPC valid only 20 days	Pattern of FTE pdf
CC rake examined at Special "A" depot	**7,500 km or 35 days**	same grace	Pattern of FTE pdf
CC rake, lower-class depot	4,500 km or 20 days		Freight Train examination pdf
Premium end-to-end (PEE)	**12 days** incl. date of issue	+3 days grace if loaded; on day 15 rake must be unloaded and offered; invalid if stabled > 24 h without offering	Pattern of FTE pdf
Intensive end-to-end (E2E)	Valid for one loaded trip to destination endorsed on BPC	examined empty; rake integrity disturbed by <= 4 wagons; empty rake must reach loading point within 4 days of BPC	both pdfs
CC rake leaving the closed circuit	treated as running on invalid BPC; returned to base depot for re-certification		Freight Train examination pdf

CAG Report 31 of 2014, para 3.2.2(f) (compliance reality): of 1,962 rakes with 6,000 km BPCs, 770 (39%) travelled 789-884 km beyond the authorised distance; of 1,772 rakes with 7,500 km BPCs, 662 (37%) ran 534-996 km beyond; MoR admitted >35% of 3,734 rakes ran 500-996 km after BPC expiry. https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf

2. Where examination is done

- CC rakes: at the **nominated base depot** where the rake was formed, on a nominated examination line jointly fixed by CRSE and CFTM; rake must not be stabled >= 24 h at an examination yard.
- Premium rakes: nominated "A" category yard chosen by CME & COM.
- Intensive E2E: nominated intensive yard, empty condition; back-loading examination at a nearby intensive yard after unloading. https://rskr.irimee.in/sites/default/files/Pattern%20of%20freight%20train%20examination.pdf

3. Time / manpower for intensive examination

Pattern	Man-hours (JPO)	Gang	Elapsed time	Source
Intensive E2E	56 man-h (Pattern pdf) / 40 man-h (FTE pdf)	2 examiners + 12 staff + welder	~4-5 h with full gang	both pdfs
Premium E2E	75 man-h	same	~6 h	both pdfs

Pattern	Man-hours (JPO)	Gang	Elapsed time	Source
CC rake	**100 man-h or 08 hours** (FTE pdf); 100 man-h (Pattern pdf); 75 man-h in one JPO item	same	**8 h** nominal	https://rskr.irimee.in/sites/default/files/Freight%20Train%20examination.pdf

The JPO sets the CC-rake examination at "100 man/hr or 08 hours" with a standard gang of two trained examiners, 12 staff and a welder.

4. Daylight rule

- CC rakes "should be examined in day light only (morning to sun set)". Night examination permitted only with minimum illumination of **100-150 lux** for under-gear examination/repair. https://rskr.irimee.in/sites/default/files/Pattern%20of%20freight%20train%20examination.pdf https://rskr.irimee.in/sites/default/files/Freight%20Train%20examination.pdf

Modelling implication: if a CC rake arrives at the base depot after ~15:00 it generally waits for the next morning; examination window ~06:00-18:00, 8 h duration, so at most one CC examination per examination line per day unless the yard is lit.

5. Sick-line and detachment detentions (for the "sick wagon" branch)

- CAG 2014 para 4.2.1: 53,815 wagons in four zones detained in sick lines for **more than 9 h** (9 h being the norm for sick-line turnout); para 4.2.2: 2,484 wagons held 48-173 days after being declared fit. https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf
- Rake integrity: an examined E2E rake may lose/replace up to 4 wagons without invalidating the BPC (JPO).

6. Suggested parameters for the model

Quantity	Value	Basis
CC BPC validity	6,000 km / 30 days (A depot); 7,500 km / 35 days (Special A)	JPO
Grace	+5 days when heading to base depot	JPO
Intensive CC examination duration	tri(6, 8, 12) h	JPO "100 man-h or 8 h"; 12 h allows for repairs/welding
Examination window	06:00-18:00 only (daylight) unless lit yard	JPO
Wait at base depot before examination (arrival->placement on exam line)	tri(0.5, 2, 10) h	CAG Table 18 arrival-to-placement ranges
Probability a wagon is detached sick at examination	model input; CAG shows 4 wagons tolerance	JPO
Sick-line repair time for a detached wagon	tri(6, 9, 24) h	CAG 9 h norm

Indian Railways norms: running-staff (loco pilot / guard) duty hours, rest, crew change

Purpose: parameters for the crew-availability constraint in the coal rake-cycle DES: maximum continuous duty, extension rule, rest at HQ/outstation, sign-on lead time and pre-departure detention.

Legal frame: Railway Servants (Hours of Work and Period of Rest) Rules, 2005 ("HOER"), under the Railways Act 1989; running staff are classified "Continuous" (54 h/week statutory, 104 h/fortnight rostered, 108 h statutory).

- IRIMEE HOER note: https://rskr.irimee.in/sites/default/files/HOER.pdf
- Ministry of Labour brief on HOER 2005: https://labour.gov.in/node/88748
- IREM Chapter 15 (Rule 1511, Breach of Rest Allowance): https://indianrailways.gov.in/railwayboard/uploads/directorate/establishment/estbl-chap15-p01-p16.htm

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1. Duty hours at a stretch

Rule	Value	Source
Running duty at a stretch "should not ordinarily exceed"	**9 h**	RBE 143/2016, Board letter E(LL)/2016/HPC/1 dated 28.11.2016 (HPC decisions) https://www.airfindia.org/2016/12/04/recommendations-of-the-high-power-committeehpc-to-review-the-duty-hours-of-running-staff-decisions-thereof-rbe-1432016/
Extension beyond 9 h	permitted only if crew given **at least 2 hours' notice** before the 9 h expire	RBE 143/2016 (same)
Total duty sign-on to sign-off	**should not exceed 11 h**	RBE 143/2016 (same)
Absolute cap "subject to severe operational exigencies"	**12 h** sign-on to sign-off	Railway Board consolidated guidelines Oct 2023 (Tribune 14.10.2023) https://www.tribuneindia.com/news/india/drivers-shouldnt-work-beyond-12-hours-railyway-board-553087
Older zonal norm (NR Operating Manual App. B, pre-2016)	10 h running duty; 8% of 23,835 crews exceeded 10 h (Dec 2014-Feb 2015)	CAG Report 47 of 2015 para 2.4.6 https://cag.gov.in/webroot/uploads/download_audit_report/2015/Union_Performance_Railways_Crew_mgmt_Report_47_2015.pdf
Fortnightly ceiling	**125 duty hours per fortnight**	RBE 120/2016, E(LL)/2016/HPC/7 dated 13.10.2016 https://www.gconnect.in/orders-in-brief/railways-orders-in-brief/decision-high-power-committee-duty-hours-running-staff.html
Continuous night duties	max **4 nights**, 4th night towards HQ	RBE 120/2016 (same)
Away from HQ	max **72 h**; >36 h monitored at division	RBE 120/2016 (same)

Rule	Value	Source
Union position (AILRSA, Aug 2025)	reiterates 9 h stretch / 11 h total; alleges CMS "fake breaks"	https://www.deccanherald.com/india/loco-pilots-union-alleges-manipulation-of-working-hours-to-hide-overutilisation-3662354

2. Rest rules

Rule	Value	Source
Rest at headquarters after duty >= 8 h	**16 h**	IREM Rule 1511 https://indianrailways.gov.in/railwayboard/uploads/directorate/establishment/estbl-chap15-p01-p16.htm
Rest at headquarters after duty < 8 h	**12 h**	IREM Rule 1511 (same)
Outstation rest after 8 h duty	**8 h**	Railway Board circular to CRIS dated 03.06.2025 (standardising to three types) https://www.deccanherald.com/amp/story/india%2Frailway-ministry-changes-outstation-rest-rules-for-train-crew-union-calls-it-unlawful-order-3576029
Outstation rest after 5-8 h duty	**6 h**	same
Outstation rest after <= 5 h duty	duty + 1 h	same
Periodic rest	4 x 30 consecutive h or 5 x 22 consecutive h per month, incl. a full night (22:00-06:00)	HOER note https://rskr.irimee.in/sites/default/files/HOER.pdf
HPC 2016 view	outstation rest below 8 h "highly inadequate"	Deccan Herald article above

3. Crew change / relief when duty expires en route

- RBE 143/2016 clause 3 covers the case where the train does not reach destination or the normal crew-changing / reliever point within the limit: the administration must arrange relief (the 2 h notice rule applies to extending the current crew). Text: https://www.airfindia.org/2016/12/04/recommendations-of-the-high-power-committeehpc-to-review-the-duty-hours-of-running-staff-decisions-thereof-rbe-1432016/
- Practice reported by CAG 47/2015 (crew management): crew changing points are nominated per division; relief crews are "called" from the running room / lobby via CMS. No Board-wide time standard for relief-crew arrival; model it as a wait on the crew lobby queue plus travel (road van) to the stabled train. https://cag.gov.in/webroot/uploads/download_audit_report/2015/Union_Performance_Railways_Crew_mgmt_Report_47_2015.pdf
- Freight trains whose crew expire en route are stabled at the nearest station until a relief crew arrives ("crew-expired detention"); this is reported in Hitavada (Dec 2024) in the context of Railway Board scrutiny of continuous night duties: https://www.thehitavada.com/Encyc/2024/12/29/Railway-Board-raises-eyebrows-over-staff-s-continuous-night-shifts.html

4. Sign-on lead time, calling, pre-departure detention (CAG Report 47 of 2015, CMS data)

Source: https://cag.gov.in/webroot/uploads/download_audit_report/2015/Union_Performance_Railways_Crew_mgmt_Report_47_2015.pdf

- Para 2.1.12: crew must sign on **10 to 45 minutes before scheduled departure** (train-ordering time). In 2,771,169 transactions 26% signed on at/after ordering time and 12% signed on < 10 min before it.
- Para 2.1.10: calling time -- 21.52% of calls made > 165 min before ordering time; 6.66% after it.
- Para 2.1.11: call acknowledgement -- 54.21% within 30 min; 16.07% >= 165 min; CMS manual standard 10 min for SMS calls.
- Para 2.1.13: sign-on approval -- 68% approved >= 30 min after sign-on.
- Para 2.1.14: sign-off approval -- 37% approved > 1 h after sign-off.
- CMS defines **Pre-Departure Detention (PDD)** = later of (sign-on, booked departure) -> actual departure. https://cms.indianrail.gov.in/CMSHelp/2.Freight/PDD.html
- CAG Report 31 of 2014 para 3.4: of 224,599 late-started goods trains at 119 points (2010-13), ~63% were late for want of locos and 18% for want of crew and path. https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf

5. Suggested parameters for the model

Quantity	Value	Basis
Max running duty before relief needed	9 h (planning); hard cap 11 h; 12 h exigency	RBE 143/2016; RB 2023
Extension allowed if notice given >= 2 h before 9 h	up to 11 h total	RBE 143/2016
Sign-on to departure (crew booking + loco/BPC check)	tri(0.5, 0.75, 2.0) h	CAG: 10-45 min sign-on lead + 68% approvals >30 min + PDD
Sign-off	tri(0.25, 0.5, 1.0) h	CAG 37% approvals > 1 h
Outstation rest before next duty	6 h (duty 5-8 h) / 8 h (duty >= 8 h)	RB 03.06.2025
HQ rest before next duty	12 h (<8 h duty) / 16 h (>= 8 h duty)	IREM 1511
Relief-crew arrival when crew expires en route	tri(1, 2.5, 6) h	no published norm; CAG late-start and detention data; treat as calibratable
Night duty cap	4 consecutive nights	RBE 120/2016

Indian Railways norms: loading / unloading free time, demurrage, wharfage

Purpose: published norms to anchor triangular (min, mode, max) distributions for the "placement -> release" activities at the coal siding (loading) and the power-house siding (unloading) in the rake-cycle DES. All figures carry a URL. Where the source is secondary (zonal web page, quiz bank, news), it is flagged.

1. Governing document

- The master document is **Rates Master Circular/Demurrage-Wharfage-Waiver/2016/0 dated 19.05.2016** (Railway Board, Traffic Commercial Dte), amended by numbered corrigenda (No. 32 of 13.09.2022 revised demurrage rates; No. 42/45 hopper-in-lieu- of-open; No. 54 of 07.11.2024 crane handling). Source (SR schedule of powers quoting the circular): https://sr.indianrailways.gov.in/cris//uploads/files/1751884201884-waier%20updated%2030062025.pdf CR Annexure to Corrigendum 54: https://cr.indianrailways.gov.in/cris/uploads/files/1731569594339-ARN%20No%20136%20of%202024%20Goods.pdf
- Free time is reckoned **from the time of placement of the rake** (Railway Board letter TC-V/2024/201/efile/3 (3462079) dated 13.05.2024, "Free time for loading/unloading of long haul rakes"): https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates-Letters/2024/Free-13_05_2024.pdf

2. Free-time table (goods sheds and private/assisted sidings)

Source: Konkan Railway "Demurrage Rules – Free Time & Charges" page, reproducing the Rates Master Circular table (zonal web page; verify against the master circular). https://konkanrailway.com/node/5508

Wagon type	Mechanised loading / unloading	Manual loading / unloading
Open (BOXN, BOXNHA, BOXNHS, BOST, BOX)	**5 h 00 / 7 h 00**	9 h 00 / 9 h 00
Hopper (BOBR, BOBRN, BOBS, BOBY)	**5 h 00 / 2 h 30**	N.A.
Flat (BFR, BRN, BFNS, BFKI)	6 h 00 / N.A.	8 h 00 / 8 h 00
Covered (BCN, BCNA, BCNHS)	N.A.	5 h (1-20 wagons), 7 h (21-30), 9 h (31+)
Container	3 h 00 / 3 h 00	
Crane-handled	7 h loading / 8 h unloading	

Historical confirmation of these magnitudes (Lok Sabha Unstarred Q 2785, answered 11.08.2005, demurrage rules effective 15.07.2005): jumbo rakes 8 h -> 9 h for loading and unloading; mechanised unloading of open wagons 5 h -> 6 h; extra free time for additional shunting/placement capped at 2 h. https://eparlib.nic.in/bitstream/123456789/516626/1/17329.pdf

Rule when both methods are used on one rake: the more restrictive (mechanised) free time applies (CR Annexure to Corrigendum 54): https://cr.indianrailways.gov.in/cris/uploads/files/1731569594339-ARN%20No%20136%20of%202024%20Goods.pdf

Engine-on-load (EOL) sidings (Rates Circular No. 21 of 2004, letter 2004/TC-I/94/214/9 Vol II dated 12.07.2004): loading all wagon types, bulk **3 h**; unloading all types except BOBRN, bulk **4 h**; unloading BOBRN **2 h**. https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/rate_inst_21-2k4.jsp

3. Coal-specific free time (loading end) -- 2024 Railway Board letters

- **Railway Board letter TC-I/2019/201/6-Part(2)(3328629) dated 22.01.2024** ("Free time for loading of rakes of Coal and iron ore"): free time for loading at goods sheds / sidings where **two or more coal/iron-ore rakes are loaded per day from a single line** was **reduced from 5 h to 3 h**.
- **Same file, letter dated 12.04.2024** ("...Power to Zone"): zonal railway may review (restore) the free time where single-line loading is below two rakes/day. Demurrage for sidings payable by the 10th of the following month. https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates-Letters/2024/free%20time%20for%20loading%20and%20unloading%20of%20rakes%20of%20Coal%20and%20Iron%20Ore%20-Power%20to%20Zone%20dated%2012_04_2024.pdf
- Bulk-online (industry forum, RLS vendor, secondary): Indian Railways "now giving 4 hours to load the entire rake", previously 10 h then 6-7 h; retractable loading chute rated up to 5,500 tph, reclaim 2,200-2,400 tph; classic Indian load-out used a 3,000 t concrete silo. https://www.bulk-online.com/en/node/60840
- Departmental quiz bank (secondary): coal demurrage slab after a **4-hour free time**: 2x base rate 0-4 h, 2.5x 4-8 h, 3x beyond 8 h. https://mycbt.in/posts/knowledge-booster-com-ldce-edition-19-jul-2026

Modelling implication: at a WCL/CIL siding loading >= 2 rakes/day the current Railway free time for loading a BOXN rake is 3 h (reduced from 5 h); 5 h still applies at low-throughput sidings. Pay-loader (manual-category) sidings historically used 9 h.

4. Loading time actuals (RLS/silo vs pay-loader) -- Coal India / press

Method	Time per rake	Source
Mechanised CHP + silo (Basundhara, MCL)	**8-10 h -> ~90 min**	BigMint analysis https://www.bigmint.co/insights/detail/indias-coal-supply-chain-strong-on-output-weak-on-delivery-747729
CCL Karo (7 MTPA) & Konar (5 MTPA) CHP+silo	"rake loading time will be reduced **from 5 hours to 1 hour**"	PIB release 2062752 (Ministry of Coal) https://www.coal.nic.in/sites/default/files/2024-10/PIB2062752.pdf
Retractable chute RLS	up to 5,500 tph (a 58-wagon, ~3,800 t rake in ~45-60 min of flow)	https://www.bulk-online.com/en/node/60840
SECL Dipka RLS (25 MTPA, commissioned 21.02.2025)	no time given	https://powerline.net.in/2025/02/26/secl-commissions-rapid-loading-system-at-dipka-megaproject/
BCCL Maheshpur RLS (5 MTPA) trial rake 20.09.2026	no time given	https://indianpsu.com/?p=41081
Truck + pay-loader, Korba	trucks wait 6-8 h to unload at siding; 150-200 truck trips per rake	https://www.bigmint.co/insights/detail/indias-coal-supply-chain-strong-on-output-weak-on-delivery-747729

CIL loaded 280.7 rakes/day overall and 255.6 rakes/day to power in FY2018-19 (Ministry of Coal Annual Report 2019-20 ch. 5): https://www.coal.gov.in/sites/default/files/2020-09/Chapter5-en.pdf

5. Unloading free time at power-house sidings

- Tariff: open wagons (BOXN) mechanised unloading (tippler) **7 h**; hopper (BOBRN) **2 h 30**; manual **9 h** (Konkan table above). EOL sidings: 4 h (open) / 2 h (BOBRN).
- Actual unloading times (BigMint, secondary): rapid unloaders / rotary tipplers **2-3 h**, mechanical grabs **6-8 h**, manual **12-15 h**; about 24 h of the lost turnaround is typically at the unloading end; agreed rake turnaround 48-72 h vs summer actual 96-120 h. https://www.bigmint.co/insights/detail/indias-coal-supply-chain-strong-on-output-weak-on-delivery-747729
- Railways (Jan 2022) said hopper-rake unloading at a plant is about **3 h per rake**, "much less" than tippler system. https://indiainfrahub.com/news/road-map-in-place-for-supplying-rakes-to-long-distance-thermal-power-stations-says-railways/
- IIMA WP 2004-09-06 (Bhatnagar, Gupta, Pappu, Dutta) on a 490 MW Indian plant: rakes of 56-58 wagons of 56-58 t (~3,300 t), ~1.5 rakes/day, 2 tipplers, free unloading time then **10 h**, observed mean unloading **14 h (sd 4 h)**, mean demurrage ~4 h/rake, waiting room 6 rakes. https://iima.ac.in/sites/default/files/rnpfiles/2004-09-06gdutta.pdf

6. Demurrage and wharfage

- Base demurrage **Rs 150 per 8-wheeled wagon per hour or part** (Corrigendum 32, letter TC-I/2022/201/efile/2(3394909) dated 13.09.2022, effective 01.10.2022). Graded: up to 6 h beyond free time base; 6-12 h +10%; 12-24 h +25%; 24-48 h +50%; 48-72 h 2x; >72 h 3x. Penal demurrage up to 6x base at congested terminals with 24 h notice (para 3.4.1). https://sr.indianrailways.gov.in/cris//uploads/files/1751884201884-waier%20updated%2030062025.pdf https://konkanrailway.com/node/5508
- Earlier rates: Rs 100/wagon/h from 2008 until raised to Rs 150 from 1 April 2013 (CAG Report 31 of 2014 para 3.6); Rs 60/bogie wagon/h under the 2004 EOL scheme. https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/rate_inst_21-2k4.jsp
- Wharfage (goods lying in railway premises after unloading) is not relevant to a siding-to-siding coal rake cycle; the Rates Master Circular covers it in Chapter II. Reckoning of free time for removal of consignment (Railway Board 24.01.2024): https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates_Master_Circulars/2024/Reckoning%20of%20free%20time%20for%20removal%20of%20consignment%20-(%20%20dated%2024_01_2024%20(i).pdf
- Demurrage waiver: 25% (Rs 267.07 cr) of accrued demurrage was waived IR-wide in 2008-13; Rs 1,056.96 cr accrued on 223,208 of 450,852 rakes (CAG 2014 para 3.6).

7. Observed "placement to release" and "arrival to placement" (CAG 31 of 2014, para 3.2.5, Table 18)

Source: https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf

Stage	Range of zonal averages 2011-13
Loading: arrival -> placement	0 h 32 m to 10 h 15 m
Loading: placement -> release (party account)	5 h 03 m to 56 h 32 m
Loading: release -> despatch	0 h 49 m to 21 h 03 m
Unloading: arrival -> placement	0 h 38 m to 15 h 07 m
Unloading: placement -> release (party account)	9 h 27 m to 124 h 12 m
Unloading: release -> despatch	1 h 28 m to 11 h 04 m

- Railway Board's envisaged total terminal detention: **16 h**; 32% of wagons at 128 points detained > 24 h; average detention during loading ranged 12-86 h and during unloading 13-141 h across points. Party free time 5-11 h by wagon type; railway account allowance 1 h each for arrival->placement and release->despatch (footnote 25).
- Terminal-yard detention norms set by zones: ECoR 1 h, ECR 8 h, WR 10 h, **CR 15 h**, ER 24 h (para 3.2.6).

8. Suggested triangular parameters (hours) for the model

Activity	min	mode	max	Basis
Loading BOXN rake, RLS/silo (incl. positioning)	1.0	1.5	3.0	CCL 1 h; MCL ~1.5 h; RB free time 3 h
Loading BOXN rake, pay-loader	4.0	6.0	9.0	RB 5 h (old free time), manual 9 h, Basundhara 8-10 h pre-silo
Placement (arrival at siding -> placed)	0.5	1.0	4.0	CAG Table 18, 1 h railway allowance
Release -> despatch (paperwork, BPC check, loco)	0.8	1.5	6.0	CAG Table 18
Unloading BOXN rake, rotary tippler x2	2.0	4.0	7.0	BigMint 2-3 h; RB free time 7 h; IIMA 14 h was 2004 era
Unloading BOBRN rake (hopper)	1.5	2.5	4.0	RB 2 h 30 free time; IR "about 3 h"
Unloading BOXN, grab/manual	6.0	9.0	15.0	BigMint 6-8 h grab, 12-15 h manual; manual free time 9 h

Lateness-history sources for trains at Chandrapur (CD) / Tadali (TAE)

Research date: 2026-10-13. Goal: obtain ~90 days of per-train, per-day delay at Chandrapur to calibrate a stochastic passenger-train arrival process for the Tadali merge.

Access notes (what could/could not be fetched from this sandbox)

Site	Fetchable?	Notes
erail.in	Yes	Timetable pages and raw `getTrains.aspx` endpoint worked. No historical delay.
etrain.info	Yes	`/train/<Name>-<No>/history` worked. Station-wise average delay with period filter.
railmitra.com	Yes	Live run for yesterday/today/tomorrow only.
indiarailinfo.com	**No** (fetch error)	Known to hold per-run arrival/departure history behind member login.
runningstatus.in	**No** (fetch error)	Page `https://runningstatus.in/history/<train_no>` exists per search results; not verified.
railyatri.in	**No** (fetch error)	`https://www.railyatri.in/train-running-history/<train_no>`; not verified.
trainman.in	**No** (fetch error)	Not verified.
confirmtkt.com	**No** (domain approval not granted)	Not verified.
goibibo.com	**No** (robots disallowed)	
direct curl to any site	**No** (egress 403)	All fetching went through the WebFetch tool.

1. etrain.info — best source actually verified (station-wise aggregate, up to 1 year)

URL pattern: `https://etrain.info/train/<Train-Name-Slug>-<train_no>/history` (other sections: `/schedule`, `/live`, `/availability`, `/map`, `/fare`; site index at `https://etrain.info/train-history`).

What it shows (verified 2026-10-13):

- For every station on the route: **average delay (minutes)** for a selectable period.
- Period selector: *last week / this month / last 3 months / last 6 months / last year* (client-side tabs, no URL parameter visible in the fetched HTML; the default render is "this month").
- Colour bar per station with counts in bands: Right Time 0-15 min, Slight Delay 15-60 min, Significant >60 min, Cancelled/Unknown. The page states that hovering/clicking a station bar reveals "detailed delay history" — i.e. a per-day list is present in the page JS but was not visible in the static text fetch. **A headless browser (Playwright/Selenium) will be needed to pull the per-day list; the "last 3 months" tab gives exactly the 90-day window.**

Sample values (period = "this month", fetched 2026-10-13):

Train	Station	Avg delay (min)	URL
12723 Telangana SF (north/UP)	Chandrapur CD	26	https://etrain.info/train/Telangana-Exp-12723/history

Train	Station	Avg delay (min)	URL
12723	Balharshah BPQ	21	same
12723	Nagpur NGP	30	same
12724 Telangana SF (south/DN)	Chandrapur CD	**86**	https://etrain.info/train/Telangana-Exp-12724/history
12724	Balharshah BPQ	48	same
12724	Nagpur NGP	63	same
61127 WR-BPQ MEMU (south/DN)	Tadali TAE	32	https://etrain.info/train/WR-BPQ-MEMU-61127/history
61127	Chandrapur CD	35	same
61127	Balharshah BPQ	-1 (recovers slack)	same

Observation useful for the model: southbound (DN) trains reach CD with much larger accumulated delay (they have come ~1000 km from Delhi) than northbound (UP) trains that just left Balharshah/Secunderabad. The DN line is the one the loaded Ghugus coal rakes must merge onto at Tadali, so DN-direction lateness variance is the one that matters most.

2. railmitra.com — single-run actuals (yesterday / today / tomorrow)

URL pattern: `https://www.railmitra.com/train-running-status/<train_no>` (also `https://www.railmitra.com/live-train-running-status/<slug>-<train_no>`). Shows scheduled vs actual arr/dep and delay per station for the current run, with Yesterday/Today/Tomorrow tabs. Depth: ~3 days only. Could be scraped daily to build a 90-day archive going forward.

Sample (run of 13-Oct-2026, train 12723): CD sched arr 12:15 / actual 12:26 (+11 min); dep 12:17 / 12:28 (+11). Other stations same run: SC +8, KZJ +32, RDM +13, BPQ +5, NGP +20, BPL +14.

3. indiarailinfo.com — richest per-run archive (needs login; not fetchable here)

URL patterns (from prior knowledge, not verified this session):

- Train page: `https://indiarailinfo.com/train/<slug>/<internal_id>` ; running-status/history tab per train.
- Station arrivals: `https://indiarailinfo.com/arrivals/chandrapur-cd/1186`, departures for Tadali `/departures/tadali-tae/4139`. Member-contributed and NTES-scraped per-run data goes back months to years, with station-wise actual times. Requires free account; fetch from this sandbox failed. Recommend a student try it manually in a browser.

4. runningstatus.in / railyatri / trainman / confirmtkt — unverified

- `https://runningstatus.in/history/<train_no>` – appeared in search results; typically shows last ~7-30 days per-run delay at destination and at a chosen station. Fetch failed here.
- `https://www.railyatri.in/train-running-history/<train_no>` – historically shows ~7 days of per-day delay with "on-time %" summary. Fetch failed here.
- `https://www.trainman.in/running-history/<train_no>` – historically 7-day per-day history. Fetch failed.
- `https://www.confirmtkt.com/train-running-status/<train_no>` – live only.

5. Official

- NTES (`https://enquiry.indianrail.gov.in/mntes/`) gives live/one-day-back only; no public history API.
- RailRadar / Where-is-my-train apps hold history but no public bulk endpoint.

Recommendation

1. Primary: **etrain.info `/history` with the "last 3 months" tab**, scraped with a headless browser for each of the ~115 trains in `timetable_tadali_chandrapur.csv`, reading the per-day popup for station `#CD` (and `#TAE` for the two MEMUs). Gives the 90-day per-day delay series directly.
2. Fallback / cross-check: indiarailinfo running-history (manual login) for the ~15 daily trains.
3. Going forward: cron-scrape railmitra "Yesterday" tab daily to accumulate an independent archive.
4. If only aggregates are obtainable, use etrain.info banded counts (0-15 / 15-60 / >60 / cancelled) per station to fit a mixture (e.g. point-mass near 0 + lognormal tail) per train and per direction.

Literature: simulation / modelling of Indian coal-rake cycles, rake turnround and rail terminals

Each entry: citation, URL, what was modelled, and the numeric parameters reported that are usable for a triangular-distribution DES of a coal rake cycle. Entries marked [no numbers] were checked and give no activity-time data in the accessible text.

A. Directly on Indian coal rakes / power-plant unloading

1. **Bhatnagar A., Gupta R., Pappu S., Dutta G. (2004). "Mathematical Modeling for Demurrage Reduction in Coal Transportation for an Indian Thermal Power Plant." IIMA Working Paper WP 2004-09-06.** PDF: https://iima.ac.in/sites/default/files/rnpfiles/2004-09-06gdutta.pdf Abstract: https://www.iima.ac.in/publication/mathematical-modeling-demurrage-reduction-coal-transportation-indian-thermal-power-plant

    - 490 MW plant, ~4,600 t/day coal, ~1.5 rakes/day; rake = 56-58 wagons x 56-58 t (~3,300 t); 3 inlet lines, 1 outlet, 2 wagon tipplers, 2 x 100 t weighbridges, waiting room 6 rakes; mine distance ~1,250 km.
    - Inter-arrival times observed 0.1-1.9 days (27 months of logbooks); arrival processes tested: deterministic, Uniform(6,14) h, Exp(mean 14 h), N(14,4), N(12,4), N(10,4) h.
    - **Unloading time per rake: mean 14 h, sd 4 h** (current); scenarios 6-14 h.
    - **Free unloading time (then) 10 h**; mean demurrage ~4.0 h/rake in all arrival scenarios (Table 4: 4.001-4.028 h); "successive detention" (queueing behind previous rake) explains R^2 = 0.85 of demurrage; Table 3 cause-wise demurrage hours per rake 1.5-12.4.
    - Use: unloading-service-time distribution shape (approx. normal, CV ~0.3) and the insight that terminal queueing, not the service itself, drives demurrage.

2. **Mukherjee A., Som A., Adak A., Raj P., Kirtania S. (2012). "Augmenting an Inbound Raw Material Handling System of a Steel Plant by Uncovering Hidden Logistics Capacity." Proc. Winter Simulation Conference 2012 (Dastur, Kolkata).** https://www.informs-sim.org/wsc12papers/includes/files/con273.pdf (also https://promodel.com/media/pdf/PR-ML-Dastur.pdf)

    - ProModel DES of an Indian integrated steel plant's raw-material unloading yard (iron ore, coal, coke, limestone rakes). One year of rake-arrival data fitted (distribution not named).
    - Table 4 mean stage times (minutes) for **coal rake to tippler**: wait arrival -> release of railway engine t0 = 365; release -> departure Yard 1 t1 = 144; travel to tippler t2 = 61; **unloading t3 = 828 (13.8 h)**; post-unloading handover t4 = 126; total T = 1,159 min (19.3 h). Iron-ore rake: t3 = 251 min, T = 348. Coke T = 410.
    - Coal tippler utilisation 26-30%; plant loco utilisation 71%; adding locos did not raise capacity (track-bound). Traditional expansion implied 14.5 h average demurrage per coal rake.

3. **Mukherjee A., Som A., Adak A. (2013). "Reducing Wagon Turnaround Times by Redesigning the Outbound Dispatch Operations of a Steel Plant." Proc. WSC 2013.** https://informs-sim.org/wsc13papers/includes/files/309.pdf

    - Loading-side DES; current wagon turnaround ~24 h vs Railway's new 6 h limit.
    - Table 2 railway-side activity ranges (min): pre-loading 25-30; wagon preparation 50-60; **train examiner (TXR) activities 50-60**; documentation 25-30; total rail-involved time <= 180 min. Loading 160-180 min with efficient cranes.
    - Breakdown of 24 h: loading 27%, weighing 32%, internal movement 13%, receipt 5%, tare weighing 4%, placement 5%, waiting for full rake 7%, documentation 7%.

4. **Shambharkar S. et al. (2015). "Simulation of Intra & Inter Yard Movement ... Indian Steel Industry." Proc. WSC 2015 (Tata Steel).** [no numbers for rake times] https://www.informs-sim.org/wsc15papers/102.pdf

5. **Manikutty S., Raghuram G., Venkata Rao V. (1992). "Simulation Models to Evaluate Railway Operating Policies." IIMA WP 1992-04-01.** https://ideas.repec.org/p/iim/iimawp/wp01096.html

    - Two IR simulations: loco assignment at a junction for iron-ore trains; and a 43 km double-line section with 5 originating/terminating stations for coal and cement, evaluating twin-single-line working, line capacity and section travel time. [parameters not in abstract]

6. **Raghuram G. et al. "A Decision Support System for Improving Railway Line Capacity." IIMA.** https://iima.ac.in/publication/decision-support-system-improving-railway-line-capacity [IR line-capacity DSS; numbers not accessible]

B. Official audits and committee reports with activity-time distributions

7. **C&AG Report No. 31 of 2014, "Management of Goods Trains in Indian Railways."** Ch. 3-4: https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_7.pdf Exec summary: https://cag.gov.in/uploads/download_audit_report/2014/Union_Performance_Railways_Management_Goods_trains_31_2014_chapter_5_Exe_Summ.pdf

    - Table 18 stage-wise detention ranges (see ir_freight_stats_and_capacity.md); 32% of wagons detained > 24 h at terminals; 16 h envisaged; speed histogram 71% / 17% / 12% for <= 20, 20-40, 40-100 km/h; BPC overrun 37-39% of CC rakes; late starts 63% loco / 18% crew+path; sick-line norm 9 h.

8. **C&AG Report No. 47 of 2015, "Crew Management in Indian Railways."** https://cag.gov.in/webroot/uploads/download_audit_report/2015/Union_Performance_Railways_Crew_mgmt_Report_47_2015.pdf

    - Sign-on 10-45 min before ordering; 26% late sign-on; 68% sign-on approvals > 30 min; 8% of crews > 10 h continuous running; call-acknowledgement distribution.

9. **Public Accounts Committee, 36th Report (18th Lok Sabha), "Punctuality and travel time in Indian Railways" (follow-up to C&AG Report 22 of 2021).** https://eparlib.sansad.in/bitstream/123456789/3017115/1/18_Public_Accounts_36.pdf

    - Goods speed 23.6 km/h (2019-20); interchange detention ranges 66-262 min at NKJ, BHC, ITR; line utilisation 137-175% on NCR/ECR sections.

10. **Standing Committee on Railways (Dec 2025), "Increasing Freight Related Earnings ... Dedicated Freight Corridors" (PRS summary).** https://prsindia.org/files/policy/policy_committee_reports/Railways_enhancing_freight_revenue.pdf

- Freight speed 23.8 km/h (2024-25); DFC 37 km/h; coal ~50% of freight earnings.

11. **Indian Railways Annual Report & Accounts 2023-24.** https://indianrailways.gov.in/railwayboard/uploads/directorate/stat_econ/2025/Indian%20Railways%20Annual%20Report%20%20Accounts%202023-24%20-English.pdf

    - WTR 5.11 days; speed 25.00 km/h; 192 wagon-km/wagon-day; 1,899 t net per train.

C. Line-capacity theory used in India

12. **Sangphong O., Siridhara S., Ratanavaraha V. (2017). "Determining Critical Rail Line Blocks and Minimum Train Headways..." Engineering Journal 21(3).** https://engj.eng.chula.ac.th/ej/article/download/1266/580

    - States Scott's formula C = 1440/(T+t) x E; 8 km block -> 10 min headway; 50 km/h freight braking distance 400 m -> 600 m minimum block.

13. **"Who is Scott?" Business Standard, 05.10.2017 (IR line-capacity practice).** https://business-standard.com/amp/article/opinion/who-is-scott-117100501278_1.html

    - 1440 - 120 maintenance = 1320 min; divide by (slowest running time + section preparation time); x 70%; 40% of 1,219 sections > 100% in 2015 White Paper.

14. **Nippon Foundation / JICA (1996), "Seminar on Railway Modernization in Myanmar", s.4 Block system for single-track sections.** https://nippon.zaidan.info/seikabutsu/1996/00501/contents/096.htm

    - N = 1440 f/(T + C); block handling C = 2.5 min (tablet), 1.5 min (automatic); f = 0.5-0.75.

15. **Shukla U. (2026). "Track Allocation in Indian Railways..." UCLA capstone; policy brief "A Corridor Pilot for Better Track-Time Governance in Indian Railways".** https://escholarship.org/uc/item/0sn0942s ; https://rosap.ntl.bts.gov/view/dot/92706 [reviews IR line-capacity and maintenance-block rules; no numbers in accessible text]

D. Industry / press sources on loading technology (secondary)

16. BigMint (2026), "India's coal supply chain: strong on output, weak on delivery." https://www.bigmint.co/insights/detail/indias-coal-supply-chain-strong-on-output-weak-on-delivery-747729
    - Silo loading 8-10 h -> ~90 min; unloading 2-3 h (rapid), 6-8 h (grab), 12-15 h (manual); agreed turnaround 48-72 h vs 96-120 h summer.
17. Ministry of Coal PIB 2062752 (Karo/Konar CHP): loading 5 h -> 1 h. https://www.coal.nic.in/sites/default/files/2024-10/PIB2062752.pdf
18. Bulk-online RLS thread: IR allows 4 h to load a rake (previously 10, then 6-7 h); 5,500 tph chute. https://www.bulk-online.com/en/node/60840

19. Indiainfrahub (Jan 2022): hopper-rake unloading ~3 h. https://indiainfrahub.com/news/road-map-in-place-for-supplying-rakes-to-long-distance-thermal-power-stations-says-railways/

E. Gaps

- No peer-reviewed Indian paper was found that publishes fitted distributions for the full coal rake cycle (loading, BPC examination, crew, en-route). The IIMA 2004 paper (unloading N(14,4) h) and the Dastur WSC 2012/2013 papers (stage means and ranges) are the closest. FOIS-based detention distributions exist only in CAG summaries (ranges and >24 h shares), not as raw histograms.

Coal rake composition and payload norms (Indian Railways)

Research date: 2026-10-13.

1. Wagon specifications (RDSO / IRIMEE / IRFCA)

Sources:

- IRIMEE (Indian Railways Institute of Mechanical & Electrical Engineering) training note "Types of wagon stock": https://rskr.irimee.in/sites/default/files/TYPES%20OF%20WAGON%20STOCK.pdf
- IRIMEE note on stainless steel / higher axle load wagons (BOXNHL design data): https://rskr.irimee.in/sites/default/files/Stainless%20steel%20wagons%2C%20Aluminium%20wagons%2C%20Higher%20Axle%20load%20wagons.pdf
- IRFCA Rolling Stock FAQ (compiles RDSO figures and revised CC+ loading): https://irfca.org/faq/faq-stock2.html
- GATX India BOXNHL product page: https://www.gatx.com/products/bogie-open-wagon-boxnhl/

Parameter	BOXN	BOXNHS	BOXNHL	BOBRN (hopper)
Tare (t)	22.47	23.1-23.2	20.6	25.6
Payload / CC per RDSO design (t)	58.81	58.08-66.18	71.0	55.68
Gross (t)	81.28	81.28-89.28	91.6	81.28
Axle load (t)	20.32	20.32-22.32	22.9	20.32
Length over couplers (mm)	10,713	10,713	10,963	10,600
Standard rake (wagons)	58	58	58	58 (IRIMEE text; 55 in one table)

Parameter	BOXN	BOXNHS	BOXNHL	BOBRN (hopper)
Payload under revised CC+8+2 loading incl. tolerance (t)	~66	~66	~70	~62 (CC+6+2)

Notes on "CC+8+2": Railway Board permits loading open wagons to Carrying Capacity + 8 t on routes cleared for 22.9 t axle load, with a further 2 t tolerance before overloading penalty (hence "CC+8+2"). For BOXN this gives 58.81 + 8 = 66.8 t nominal, ~66 t practical (IRFCA "revised payload incl. tolerance" = 66 t; gross 88.47 t, rake 5,233 t gross). BOBRN on CC+6+2 routes: ~62 t. BOXNHL already designed for 22.9 t axle load: 70-71 t. (IRFCA notes BOXNM1/BOXNHSM1 22.9 t variants give 5,399 t gross per rake under CC+8+2.)

2. Rake composition

- Standard IR coal block rake = **58 BOXN-family wagons** (+ brake van historically; most freight now runs with end-of-train telemetry, and 59-wagon formations are common when a spare wagon is attached). IRIMEE and IRFCA both state 58 wagons/rake for BOXN, BOXNHS, BOXNHL; some BOBRN rakes are 58-59.
- Central Railway Nagpur Division record day 5-Nov-2020: 42 coal rakes = 2,448 wagons => **58.3 wagons/rake** (https://news.railanalysis.com/?p=37597).
- Central Railway Highlights 2024-25: WCL loading fell by 536 rakes = 2.14 MT => **~3,990 t/rake** on CR coal (https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf). Goods-shed coal: 484 rakes = 1.93 MT => ~3,990 t/rake also.

3. Tonnes of coal per rake (payload, not gross)

Formation	t/wagon	t/rake
58 x BOXN at design CC	58.8	3,411 (IRFCA cites 3,411 t for an "ordinary BOXN rake")
58 x BOXN at CC+8 (+2 tol.)	64-66	3,700-3,830
59 x BOXN at CC+8	64-66	3,780-3,900
58 x BOXNHL at CC (70-71)	70-71	4,060-4,120
59 x BOXNHL	70-71	4,130-4,190
58 x BOBRN at CC+6	60-62	3,480-3,600

The user's prior (58/59 wagons, ~58-64 t/wagon, 3,800-4,000 t/rake) is confirmed: it corresponds to BOXN-family rakes loaded under CC+8+2. The CR 2024-25 aggregate (~3,990 t/rake) sits at the upper end because the WCL fleet is now largely BOXNHL. For the CSTPS simulation use **~3,900 t/rake (range 3,700-4,150)**, 58-59 wagons.

Conversion to CEA daily receipt: CSTPS receipt of 29.9-47.5 kt/day ('000 T in CEA report) = 7.7-12.2 rake- equivalents/day of all modes; after deducting ~7 kt/day of ropeway/pipe-conveyor coal, rail arrivals are about 6-10 rakes/day typically, up to ~11-12 at high PLF.

4. Wagon-type implications for unloading

- BOXN/BOXNHL/BOXNHS (open, top-loaded) need wagon tipplers (or manual/payloader) => 7 h free time.
- BOBRN (bottom-discharge hopper) discharge over a track hopper => 2.5 h free time; CSTPS has track hoppers at CHP-B and CHP-C, so some WCL rakes may arrive as BOBRN.
- Loaded BOXNHL rake gross ~5,300 t; WAG-9/WAG-7 single-headed on the Wardha-Ballarshah section.

Section capacity notes: Wardha–Chandrapur–Balharshah (CR Nagpur Division) and the Tadali–Ghugus spur

Research date: 2026-10-13. All figures are quoted from the cited public sources; nothing here is from an official Working Time Table (WTT), which is not published online for CR Nagpur division.

1. Passenger-train traffic profile at Chandrapur (derived from `timetable_tadali_chandrapur.csv`)

Source: erail.in trains-between-stations pages and raw endpoint (URLs in the CSV). 115 distinct scheduled trains (58 southbound/DN toward Balharshah, 57 northbound/UP toward Wardha/Nagpur). Weighted by days-of-run this is **146 DN + 139 UP = 285 passenger-train passings per week, i.e. about 40.7 per day (≈20.9 DN, ≈19.9 UP).** Only 15 DN and 14 UP trains are daily; the rest are weekly/bi-weekly, so the actual count on a given weekday varies roughly 17–24 per direction.

Only two trains stop at Tadali (TAE): 61127 WR–BPQ MEMU (TAE dep 09:27, CD arr 09:52, DN) and 61128 BPQ–WR MEMU (CD dep 18:53, TAE arr 19:12, UP). Everything else runs through Tadali non-stop; CD timing minus ~12–15 min (DN) or plus ~15–19 min (UP) gives the Tadali passing time (CD–TAE is 15 km per erail).

Hour-of-day histogram (average passings per day = weekly count / 7; "daily" = number of 7-day trains in that hour):

Hour	DN (south) /day	DN daily trains	UP (north) /day	UP daily trains	Both /day
00	0.43	0	1.00	1	1.43
01	0.43	0	0.00	0	0.43
02	0.14	0	0.00	0	0.14
03	0.00	0	1.00	0	1.00
04	0.14	0	0.43	0	0.57

Hour	DN (south) /day	DN daily trains	UP (north) /day	UP daily trains	Both /day
05	0.00	0	1.43	1	1.43
06	1.14	0	1.57	1	2.71
07	1.29	0	0.71	0	2.00
08	1.00	0	1.71	1	2.71
09	2.00	2	0.71	0	2.71
10	0.14	0	1.00	0	1.14
11	1.00	0	1.00	1	2.00
12	1.57	1	1.29	1	2.86
13	**4.14**	4	1.14	0	**5.29**
14	1.00	1	0.00	0	1.00
15	1.00	1	1.00	1	2.00
16	0.86	0	0.14	0	1.00
17	0.71	0	1.29	0	2.00
18	1.29	1	2.00	2	3.29
19	1.71	1	0.00	0	1.71
20	0.29	0	0.00	0	0.29
21	0.43	0	0.00	0	0.43
22	0.00	0	1.71	1	1.71
23	0.14	0	0.71	0	0.86

Quiet windows (likely freight paths):

- **DN line (the one loaded Ghugus rakes merge onto at Tadali, heading to CD/CSTPS):**
  - **20:30 – 06:20** is almost empty: only 12670 (21:07, Tu/Th), 22614 (21:47, Th), 04109 (23:59, Sa), 12772 (00:12, W/F/Su), 22664/22738 (01:10), 01925 (02:22, Th), 04171 (04:22, Su). No daily train at all between 19:35 (12722 Dakshin) and 09:38 (12724 Telangana).
  - **10:00 – 10:55** (only 22109 on Wednesdays), **14:16 – 15:04**, **15:09 – 16:04**.
  - **Peak to avoid: 12:58 – 13:41** (five daily trains 20806, 12616, 12655, 12792 plus weekly 22151 in 43 min) and 11:36–12:20.
- **UP line (empties returning Chandrapur → Tadali → Ghugus):**
  - **19:00 – 22:00** empty; **23:35 – 03:20** empty except 12656 (00:27); **13:58 – 15:24**; **15:28 – 17:29**.

    - Peaks: 03:25–03:42 (up to 5 weekly trains bunch), 06:00–06:32, 07:15–08:27, 10:15–10:45, 12:45–13:02, 17:30–18:53.
- Both lines quiet together: **19:50 – 22:00 (DN)** / **20:30 – 22:00**, and **01:15 – 03:20**.

Caveat on days-of-run: the 7-bit MTWTFSS string is what erail returns for the CD→X query. erail normally shows the running day at the "from" station, but for long-distance trains that reach CD on day 2/3 this should be verified train-by-train before being used to generate a specific calendar day.

2. Line capacity utilisation and traffic (news / official)

- **Wardha–Ballarshah line-capacity utilisation: 145 % (UP) and 163 % (DN)**; Itarsi–Nagpur 151 %/154 %. Goods trains average ~10 km/h on the corridor (including detentions). Third line "nearly 60 % complete" (Jan 2023); 4th line (88 km Wardha–Ballarshah) at survey stage; traffic expected to grow ~20 % after 3rd line. Source: The Hitavada, 28 Jan 2023, https://thehitavada.com/Encyc/2023/1/28/Indian-Railways-plans-4th-corridor-on-Itarsi-Ngp-Wardha-Ballarshah-routes.html
- Wardha–Ballarshah 3rd line: 160 km, Rs 1,035 cr, 25 % done on 132 km section (Mar 2022); Sewagram–Sonegaon chord commissioned. Source: The Hitavada, 12 Mar 2022, https://www.thehitavada.com/Encyc/2022/3/12/No-solid-progress-in-Rly-projects-aimed-at-decongesting-clogged-network.html (Sewagram–Sonegaon section inauguration: https://indianinfrastructure.com/2019/09/18/mahrashtra-government-inaugurated-sewagram-sonegaon-section-of-the-wardha-sewagram-ballarshah-third-line-project/)
- Ballarshah interchange with SCR: "about 50 trains criss-cross daily", average interchange detention 72 min vs SCR and ~207 min vs SECR; ~5.5 rakes/day via Pimpalkuti crossover; 3rd line Wardha–Ballarshah "under progress", 4th line DPR being prepared (Nov 2025). Source: The Hitavada, 4 Nov 2025, https://www.thehitavada.com/Encyc/2025/11/4/eyeing-3000-mt-loading-rlys-plans-to-end-detention-at-inter-changes.html
- **CR Nagpur Division "About Us" (Dec 2023)**: 993 route-km; ~146 Mail/Express+Passenger trains/day division-wide (elsewhere in same doc 110.6 M/E + 14 passenger); **~225 freight trains interchanged per day**; 44.39 MT loaded in 2022-23 of which **~90 % coal**; 7,142 coal rakes Apr–Dec 2023 (≈26/day division-wide); WCL loading points include **Ballarshah, Babupeth, Chandrapur, Majri, Ghugus, Wani**; private coal loading at **PVIT (Vimla siding, Tadali)**; coal unloading at **CSTPS Chandrapur**, Warora, Tadali and Sindi private plants; POL unloading at Tadali; ACC cement siding at Ghugus. Third-line works commissioned 2023: SEGM–CHJC (Sewagram–Chitoda-Hinganghat area), Narkher–Teegaon, Khapri–Butibori 3rd/4th. Source: https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf
- **Tadali–Ghugus doubling sanctioned (Jul 2026)**: 14 km, currently **single line**, Rs 228 cr, freight-oriented (coal, iron ore, gypsum, POL, sand, laterite ash), +4.5 MTPA expected. No commissioning date. Sources: https://www.itln.in/railway/indian-railways-doubles-tadalighugus-line-for-228-cr-1359929 ; https://indianinfrastructure.com/2026/07/22/ir-approves-tadali-ghugus-doubling-rail-line-project-in-maharashtra/ ; https://www.nbmcw.com/news/metro-railways/indian-railways-approves-doubling-of-tadali-ghugus-railway-section-in-maharashtra-at-rs-228-cr.html
- Ballarshah–Kazipet 3rd line (south of BPQ, SCR): Asifabad–Sirpur Kaghaznagar section completed Jul 2026. Source: https://indianinfrastructure.com/2026/07/27/work-completed-on-asifabad-sirpur-kaghaznagar-section-under-ballarshah-kazipet-third-line-project-in-telangana/

3. Station layouts (platforms / loops)

- **Chandrapur (CD)**: 3 platforms, **6 tracks** (so ~4 non-platform/goods loops incl. CSTPS siding takeoff); ~105 passenger trains stopped daily (2018 figure); electrified 1988-89. Source: https://en.wikipedia.org/wiki/Chandrapur_railway_station
- **Balharshah (BPQ)**: 5 platforms, 8 tracks. Source: https://en.wikipedia.org/wiki/Balharshah_railway_station
- **Tadali (TAE)**: no public platform/loop count found (ixigo, railmitra, nativeplanet pages carry no layout data: https://www.ixigo.com/train-stations/tadali-tae-railway-station , https://www.railmitra.com/stations/Tadali-TAE). It is the junction for the single-line Ghugus branch and hosts PVIT (Vimla) coal siding and a POL siding (CR About Us). For the simulation assume: 2 main lines + at least 1 branch reception/dispatch loop + siding lines; verify from Google Earth / OpenRailwayMap (https://www.openrailwaymap.org) before fixing the parameter.
- Nagpur division also lists "Talni converted to B class with two Long Haul Loops" (2023), indicating the division is adding long-haul (2-rake) loops on the corridor. Source: CR About Us Dec 2023 (above).

4. Implications for the model

- Treat the Wardha–Balharshah double line as saturated (>140 % chart utilisation): freight paths exist only in the gaps between the ~20 passenger passings/direction/day plus other freight (the division interchanges ~225 freight/day; the Nagpur–Ballarshah trunk carries the bulk of CR's coal traffic).
- DN-direction scheduled passenger gaps at Tadali are longest 20:30–06:20 (night) and 14:15–16:00; the mid-day 12:55–13:45 bunching is the worst slot for a loaded rake to merge.
- Lateness is large and asymmetric: DN long-distance trains arrive at CD with ~25–90 min average delay (etrain.info, Oct 2026), so scheduled gaps shift substantially; use the 90-day per-day series (see lateness_sources.md) rather than the WTT gaps alone.
- Freight occupancy on the merge: a loaded ~59-BOXN rake at 30–40 km/h over the 15 km TAE→CD takes 25–30 min of block occupancy, versus ~10–12 min for an express; headway with intermediate block signals assumed ~8–10 min.

CSTPS coal unloading facilities, free time and demurrage rules

Research date: 2026-10-13.

1. Unloading facilities at CSTPS (from MAHAGENCO / CPRI documents)

Primary source: CPRI vetting report on MAHAGENCO CSTPS coal handling plants (filed with MERC data-gap replies) https://www.mahagenco.in/uploade/Petition/Data%20Gaps%20Set-I%20Annexures/DG%20-%20026%20_Vetting%20Reports/CPRI%20Vetting%20REPORT.pdf

CHP	Units served	Receiving equipment	Notes
CHP-A	3-4 (2x210 MW)	Rail, aerial ropeway (1.2 km from WCL Durgapur mine, ~7,000 t/day), road	daily need ~8,000 t
CHP-B	5-7 (3x500 MW)	**2 wagon tipplers** (L&T, commissioned 1992) + **track hopper** for BOXN and BOBR(N) wagons; 2 stacker-reclaimers (Elecon, 1992); 22 km of conveyors, 33 conveyor streams	daily need ~24,000 t
CHP-C	shared	**Track hopper** (single conveyor stream) for BOBR/UTS bottom-discharge wagons; road unloading; closed pipe conveyor from WCL Bhatadi OCM	—
CHP-D	8-9 (2x500 MW)	**3 wagon tipplers** (Elecon, commissioned 2016)	MAHAGENCO tender Jul-2026 for strengthening end frames of a CHP-D wagon tippler (tender ID 3000070885, https://tender18.com/tenders-details/16552146)

Total: **5 wagon tipplers + 2 track-hopper lines**, i.e. up to 5 independent BOXN unloading streams plus the BOBRN hopper route. Rated capacities are not stated in the CPRI document. Industry norms (for sizing the simulation):

- Rotaside / rota-tippler for BOXN: typically rated 20-25 tips/hour (one tipple = one wagon of ~60-65 t), i.e. 1,200-1,500 t/h each; a 58-59 wagon rake takes ~2.5-3 h on one tippler at rated speed, 4-6 h in practice with shunting (side-arm charger, placement of cuts of ~10-20 wagons). Source for typical tippler rating: NTPC CHP technical specs, e.g. Lara STPP 2x800 MW spec https://ntpctender.ntpc.co.in/uploads/tech_job_15486.html and Darlipali https://ntpctender.ntpc.co.in/uploads/tech_job_15561.html (tippler systems specified at ~20 tips/h, 1,500 t/h conveyors, track hopper at 1,500-4,000 t/h).
- Track hopper (bottom-discharge BOBRN rake): whole rake discharges in ~30-60 min at walking speed; free time 2.5 h.

No merry-go-round (MGR) system at CSTPS; the plant is fed by IR rakes placed into the power-house siding at Chandrapur, plus ropeway and pipe conveyor from the adjacent WCL Durgapur/Bhatadi/Padmapur mines.

2. Indian Railways free time for unloading (official figures)

Governing document: Railway Board "Rates Master Circular/Demurrage-Wharfage-Waiver/2016/0" (Chapter I, paras 2.2-2.5) with corrigenda up to No. 54 (07-11-2024). The zonal railways republish the tables; the Central Railway goods tariff ARN No. 136 of 2024 reproduces the current table: https://cr.indianrailways.gov.in/cris/uploads/files/1731569594339-ARN%20No%20136%20of%202024%20Goods.pdf A clean HTML version of the same table is on Konkan Railway's site (same Board rules apply IR-wide): https://konkanrailway.com/node/5508

Free time at railway terminals and private/assisted sidings (non-steel-plant, non-EOL), per rake:

Wagon type	Mechanised loading	Mechanised unloading	Manual loading	Manual unloading
Open wagons (BOXN, BOXNHL, BOXNHS, BOST, BOX…)	5 h	**7 h**	9 h	**9 h**

Wagon type	Mechanised loading	Mechanised unloading	Manual loading	Manual unloading
Hopper wagons (BOBR, BOBRN, BOBY, BOBS)	5 h	**2 h 30 min**	n.a.	n.a.
Flat (BRN, BFNS…)	6 h	n.a.	8 h	8 h
Containers	3 h	3 h	—	—

So for CSTPS:

- BOXN/BOXNHL rake unloaded on wagon tipplers = "mechanised unloading of open wagons" = **7 hours free time**.
- BOBRN rake on track hopper = **2.5 hours free time**.
- A manual (payloader/road) unloading of BOXN = 9 h.
- Engine-on-load (EOL) terminals (para 2.5): "others" 3 h; steel-plant open-wagon unloading 5 h. Power-house sidings are not EOL by default; CSTPS placement is by the siding shunting engine, so the 7 h figure applies.
- Free time is reckoned from the time of placement (or from "placement advice"/expiry of 2 h after rake arrival at the serving station if the siding cannot accept, per Rates Master Circular). Railway Board letter dated 24-01-2024 on reckoning of free time: https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates_Master_Circulars/2024/Reckoning%20of%20free%20time%20for%20removal%20of%20consignment%20-(%20%20dated%2024_01_2024%20(i).pdf
- Coal/iron-ore LOADING sidings: free time cut from 5 h to 3 h where 2 or more rakes are loaded per day from a single line; zonal railways empowered to apply/review (Railway Board letters TC-I/2019/201/6-Part(2) dated 22-01-2024 and 12-04-2024): https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates-Letters/2024/free%20time%20for%20loading%20and%20unloading%20of%20rakes%20of%20Coal%20and%20Iron%20Ore%20-Power%20to%20Zone%20dated%2012_04_2024.pdf Relevant for the WCL Ghugus/Wani sidings in the model: use **3 h loading free time** at busy WCL sidings.
- Ancillary circulars: SECR Rates Circular 234(G)/2022 (Corrigendum 32 dated 13-09-2022, waiver guidelines effective 01-10-2022) https://secr.indianrailways.gov.in/uploads/files/1667385128436-sign%20rc%20234_rc%20234_merged.pdf ; Board letters on waiver/refund of demurrage 14-08-2024 and 2025.

3. Demurrage charges (same source: Rates Master Circular 2016 + Konkan/CR tariff pages)

- Base rate: **Rs 150 per 8-wheeled wagon per hour** (or part) of detention beyond free time. (Rate in force since the 2017 revision; private-owned wagons attract a lower/nil rate.)
- Escalation with excess detention (slab applies to whole period): | Detention beyond free time | Rate | |---|---| | up to 6 h | Rs 150 /wagon/h | | 6-12 h | Rs 165 (base +10%) | | 12-24 h | Rs 187.50 (base +25%) | | 24-48 h | Rs 225 (base +50%) | | 48-72 h | Rs 300 (2x) | | > 72 h | Rs 450 (3x) |
- For a 58-wagon rake: Rs 8,700 per hour for the first 6 h of overstay; a 12-hour overstay costs 58 x 12 x 165 = Rs 1,14,840; a full day's overstay ~Rs 2.6 lakh.
- Demurrage at sidings is billed monthly, payable by the 10th of the following month (Board letter 12-04-2024 refused to extend to the 20th).
- Waiver: zonal CCMs may waive up to defined limits; requests judged by the policy in force on the date of completion of unloading (Corrigendum 32/2022).

4. Simulation parameters suggested from the above

- Unloading servers at CSTPS: 5 tipplers (2 at CHP-B, 3 at CHP-D) + 1-2 track-hopper lines for BOBRN.
- Service time per BOXN rake on a tippler: triangular(3, 4.5, 7) h; free time 7 h; demurrage per rake-hour beyond free time = 58 x 150 = Rs 8,700 (escalating as above).
- Service time per BOBRN rake on track hopper: 0.75-1.5 h; free time 2.5 h.
- Loading at WCL siding: free time 3 h (busy sidings) / 5 h (others), same demurrage scale.

WCL coal loading at Ghugus and Nagpur-division coal rake flows

Research date: 2026-10-13. Items marked **[NOT VERIFIED]** could not be sourced.

1. Sidings served from Ghugus (GGS)

Siding	Code	Details	Source
Ghugus Colliery Siding (WCL)	**GSG**	4 km from GGS; listed as both Independent and Non-Independent in erail's siding table	https://erail.in/stations/GGS ; https://vcrc.in/wp-content/uploads/2020/05/List_Coal_Siding_090719.pdf
Associated Cement Co. (ACC) Siding, Ghugus	**CESG**	2 km from GGS; cement loading point; also loads coal (CR Highlights mention "CESG coal loading" fell in 2024-25)	https://erail.in/stations/GGS ; https://cr.indianrailways.gov.in/cris//uploads/files/1740559903393-high%20lights.pdf
WCL "Ghugus Old Siding"	—	WCL Marketing & Sales notice: rakes for FSA consumers shifted from Umrer Siding (G-11) to **Ghugus Old Siding (G-10 grade)**; 41 pending rakes listed Sep-2024 to Nov-2025 (consumers: Bilt Graphics Paper 13, Indrajit Power 7, UltraTech Manikgarh 7, West Coast Paper 5, Atul 3, ACC 2, Vardhman 2, Ambuja 1, JK Paper 1). Implies at least an "Old" and presumably a "New" siding at Ghugus.	https://docs.westerncoal.in/strapi/uploads/List_Of_Rakes_54fdf97d57.pdf
Ghugus OCP (WCL Wani Area)	—	MoEF compliance doc (Mar 2024): Ghugus OC dispatches via the "Old Railway Siding" (14 rain guns, CTO valid to 31.03.2027) and two Coal Handling Plants; tippers from Nakoda OC and other Wani Area mines also use the siding/CHP. Ghugus OC itself: "Coal exhausted and no mining activity is going on" (mine closure phase). Loading method not stated.	https://docs.westerncoal.in/Ghugus%20OC_3-2024.pdf

WCL area structure: Ghugus is in **WCL Wani Area** (the Ghugus OCP doc is titled "Ghugus Opencast Project, WCL Wani Area"); copy-list of the rakes notice names Ballarpur, Wani and Wani North Areas. Padmapur and Naigaon are **not** mentioned in any Ghugus document found; Padmapur belongs to WCL Chandrapur Area (loads at Chandrapur/Babupeth), not Ghugus. **[Padmapur/Naigaon link to Ghugus NOT VERIFIED]**

Loading method at Ghugus: silo/RLS **not found**; the documents refer to CHPs and "wagon loading points", consistent with pay-loader/CHP wharf-wall loading. **[NOT VERIFIED]**

2. Rakes/day from Ghugus siding

Figure	Value	Source
Ghugus siding dispatch, Dec 2018	Umrer and Ghugus sidings each rose from **4–5 to 7–8 rakes/day**, plan **9–10 rakes/day** (Jan 2019 "RailCoal Week"); crossing station commissioned in Umrer Area and "a similar facility at Wani Area"	https://www.business-standard.com/article/pti-stories/wcl-to-augment-coal-despatches-from-mines-118122900352_1.html
WCL ad-hoc rakes via Ghugus Old Siding	41 rakes over Sep-24 to Nov-25 under NRS linkage auction (peak 8 rakes in Oct-25) — these are additional to FSA/power rakes	https://docs.westerncoal.in/strapi/uploads/List_Of_Rakes_54fdf97d57.pdf
Current (2025-26) Ghugus rakes/day	**Not published.** Best modelling range: 6–10 loaded rakes/day from GSG plus ~1 cement/coal rake/day from CESG. **[ASSUMPTION]**	—

3. Nagpur division coal loading (CR)

Figure	Value	Source
WCL coal loading points in Nagpur division	Ballarshah, Babupeth, Chandrapur, Majri Chargaon, **Ghugus**, Umrer, Wani, Parasia, Hirdagarh, Palachauri; other coal points: PVIT (Vimla siding, Tadali), Chikni, Wani Goods Shed, MKCW siding	https://cr.indianrailways.gov.in/cris//uploads/files/1705659988988-About%20us%20Dec23.pdf
Nagpur division originating loading	44.39 MT in 2022-23 (>54 % of CR); coal ≈ 90 % of loading; 7,142 coal rakes Apr–Dec 2023 (≈ **26 coal rakes/day**)	same
Nagpur division 2024-25	45.47 MT vs 50.05 MT in 2023-24 (−9.1 %); WCL coal loading fell 1.5 rakes/day (536 rakes, 2.14 MT → **≈3,990 t/rake**); goods-shed coal fell 1.3 rakes/day (484 rakes, 1.93 MT) after RCR ban	https://cr.indianrailways.gov.in/cris//uploads/files/1753868156096-Highlights%202024-25.pdf
Nagpur division FY25 coal rakes (to Jan-25)	6,723 coal rakes earning Rs 2,718 crore; interchange 228.6 trains/day	https://cr.indianrailways.gov.in/cris//uploads/files/1740559903393-high%20lights.pdf
Nagpur division May 2024	1,142 rakes loaded in month (≈ **37 rakes/day total**), of which coal 910 rakes (≈ **29 coal rakes/day**), Rs 386.02 crore coal earnings	https://indiaseatradenews.com/central-railways-nagpur-division-records-freight-earnings-of-rs-427-cr-for-may/
Nagpur division April 2026	3.97 MT loaded (CR total 6.72 MT; CR 2,582 rakes in month)	https://indianinfrastructure.com/2026/05/14/central-railway-achieves-freight-loading-of-6-72-mt-during-april-2026/
Division target/goal	45 MT target 2024-25; long-term 90 MT originating; coal >75 % of receipts; iron ore ≤2 rakes/day	https://www.thehitavada.com/Encyc/2025/1/25/Rs-100-crore-works-ayed-for-upgradation-of-goods-shed.html

Verification of "Nagpur division loads 100+ rakes/day": NOT supported. Published figures imply ~37 total rakes/day and ~26–29 coal rakes/day originating in Nagpur division (2023-25). The 100+ figure likely conflates originating loading with the ~228 trains/day interchange.

4. WCL company-wide rail dispatch

Figure	Value	Source
WCL 2019-20	57.6 MT produced, 52.5 MT dispatched; rail average 23 rakes/day, peak 29; ~90 % of rail dispatch via Central Railway	https://thehitavada.com/Encyc/2020/9/11/Coal-supply-WCL-to-double-rail-despatch.html ; https://www.maritimegateway.com/western-coalfields-ltd-almost-double-coal-despatch-capacity/
WCL 2020-21 "Mission 100 Days"	target 40 rakes/day avg, 50 peak (43–44 CR, 4–5 SECR, 2 SCR); record **50 rakes in a day on 5 Nov 2020** (CR 42, SECR 6, SCR 2); 2.46 lakh t/day record dispatch; target 65 rakes/day by Dec 2020	https://www.thehitavada.com/Encyc/2020/11/9/WCL-dispatches-50-rakes-creates-history.html
WCL Dec 2018	28–29 rakes/day achieved, 32/day planned from 1 Jan 2019 (CR 28–29, SECR ~3, SCR 1); rail >60 % of dispatch; 1.25 lakh t/day by rail target	https://www.business-standard.com/article/pti-stories/wcl-to-augment-coal-despatches-from-mines-118122900352_1.html
WCL FY 2023-24	Production 69.11 MT; dispatch 70.20 MT; power sector 60.05 MT; **rail-mode dispatch 39.16 MT** (record, +16.4 %) ⇒ ≈ 39.16e6/3,990/365 ≈ **27 rakes/day**	https://psuwatch.com/amp/story/hindi-news/wcl-western-coalfields-limited-coal-production-record-fy24
WCL FY 2024-25	Production flat at ~69 MT (Business Standard, Jan 2025)	https://www.business-standard.com/companies/news/western-coalfields-faces-flat-coal-production-at-69-mt-amid-challenges-125011900116_1.html
CIL modal split (Jan–Nov 2024)	Rail 55 %, road 31 %, MGR 13 %, conveyor/rope 2 %	https://coal.gov.in/sites/default/files/2025-02/chap10AnnualReport2025en2.pdf

5. Rake allotment / indent system

Item	Value	Source
Indenting	Customers register rake demand at goods office or electronically via **FOIS e-Demand (e-RD)**; Wagon Registration Fee Rs 50,000 per rake, adjustable against freight; indent position/outstanding indents trackable in FOIS	https://fois.indianrail.gov.in/RailSAHAY/pages/FAQs.jsp
e-RD module changes 2024	Railway Board letter "Modifications in e-RD module of FOIS web portal" (2024)	https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates-Letters/2024/Modifications%20in%20e-RD%20module%20of%20FOIS%20web%20portal.pdf
Rate/registration circulars	Goods Tariff Chapter II (Mar 2024) and registration ("Registy") letter 2024	https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates_Master_Circulars/2024/Chapter%20II%20Goods%20Tariff_11_03_2024.pdf ; https://indianrailways.gov.in/railwayboard/uploads/directorate/traffic_comm/Rates-Letters/2024/Registy.pdf
Demurrage	Rs 150 per wagon per hour (or part); free time varies by wagon type and terminal (not quantified on page); siding charges based on engine-hour cost and trip time serving station ↔ siding	https://fois.indianrail.gov.in/RailSAHAY/pages/FAQs.jsp
Coal-specific allotment	WCL rakes move under FSA linkages, NRS linkage auction and spot e-auction; WCL publishes rake seniority lists per siding (e.g., Ghugus Old Siding list)	https://docs.westerncoal.in/strapi/uploads/List_Of_Rakes_54fdf97d57.pdf

6. Suggested model inputs

- Loaded rake ≈ 3,990 t (58–59 BOXN wagons); empties placed on indent, loading at CHP/wharf wall.
- Ghugus demand: 6–10 coal rakes/day (GSG) + ~1 rake/day CESG; sensitivity up to 13 (doubling target +3/day).
- Loading time per rake at a non-silo WCL siding: not published; use CR free time for siding loading (typically a few hours) as a parameter **[ASSUMPTION]**.